# AI POWERED KITCHEN ECOSYSTEM FOR FOOD WASTE REDUCTION AND NUTRITION OPTIMIZATION

Project Id: 25-26J-351

Final SummaryReport

Muraleswaran D

IT22339010

BSc (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

April 2026

# PERSONALIZED AI FOOD EXPIRY PREDICTOR

Project Id: 25-26J-351

Final Summary Report

BSc (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

April 2026

## DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

| Name | Student ID | Signature |
|---|---|---|
| Muraleswaran D | IT22339010 | |

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

…….. ……

Signature of the supervisor Date

## ABSTRACT

Food waste is a major global concern driven by inaccurate shelf-life estimation and reliance on static expiration labels that ignore real-world storage conditions and user behavior. This paper presents a hybrid intelligent system for personalized food expiry prediction based on an Adaptive Expiry Intelligence Framework . The system integrates a CatBoost regression model with rule-based biological safety validation and adaptive user feedback mechanisms. The model predicts remaining shelf life using features such as food type, storage conditions, temperature, and humidity. A safety layer enforces biologically valid bounds and manufacturer constraints, while an Adaptive Expiry Adjustment module continuously refines predictions based on user feedback. Additionally, a Smart Consumption Priority mechanism ranks food items by urgency to support timely consumption. Experimental results show that the proposed approach achieves an R squared score of 0.7443 and a mean absolute error of 0.096 days, outperforming baseline models. The system provides a practical solution for reducing food waste and improving household food management.



***Keywords:*** *Food waste, Expiry prediction, Personalized Systems, Machine learning, Hybrid Intelligent Systems, Consumption patterns, Storage methods, Sustainability, Decision Support Systems*

# Contents

DECLARATION ..... iii

ABSTRACT ..... iv

LIST OF FIGURES ..... vii

LIST OF ABBREVIATIONS ..... x

1.1 Background & Literature Survey ..... 4

1.2 Research Gap ..... 8

1.3 Research Problem ..... 10

2. OBJECTIVES ..... 12

2.1 Main Objectives ..... 12

2.2 Specific Objectives ..... 12

3. METHODOLOGY ..... 13

3.1 Methodology ..... 13

3.1.1 System Overview ..... 13

3.1.2 System Workflow ..... 15

3.1.3 Data Featuring ..... 19

3.1.4 Data preprocessing ..... 20

3.1.5 Feature Engineering ..... 21

3.1.6 Adaptive Expiry Intelligence Feature ..... 23

3.1.6.1 Conceptual Foundation and Motivation ..... 23

3.1.6.2 Layered Architecture and Operational Mechanism ..... 24

3.1.6.3 Technical Advantages and Research Contribution ..... 26

3.1.7 Adaptive Expiry Drift ..... 27

3.1.7.1 Concept and Motivation of AED ..... 27

3.1.7.2 Operational Mechanism and Learning Process ..... 28

3.1.7.3 Integration with AEIF and System Workflow ..... 29

3.1.7.4 Technical Advantages and Contribution ..... 30

3.1.8 Smart Consumption Priority ..... 30

3.1.8.1 Concept and Motivation of SCP ..... 30

3.1.8.2 Priority Scoring Mechanism ..... 31

3.1.8.3 Integration with System Workflow ..... 33

3.1.8.4 Technical Advantages and Contribution ..... 34
3.2 Tools and Technologies ..... 34
3.4.1 System Implementation ..... 39
3.4.1.1 Model Loading and Prediction Execution ..... 41
3.4.1.2 Feature Engineering and Input Construction ..... 42
3.4.1.3 AEIF Safety Validation ..... 42
3.4.1.4 Personalization Activation (AED Logic) ..... 43
3.4.1.5 Prediction Pipeline Integration (Backend Flow) ..... 43
3.4.1.6 Data Persistence and History Tracking ..... 44
3.4.2 Testing ..... 47
3.4.2.1 Functional Testing ..... 47
3.4.2.2 API and Integration Testing ..... 48
3.4.2.3 Scenario-Based Testing ..... 48
3.4.2.4 Edge Case Testing ..... 49
3.4.2.5 User Interface Validation ..... 49
4 RESULTS AND DISCUSSION ..... 49
4.1 Results ..... 49
4.2 Research Findings ..... 54
4.3 Model Evaluation ..... 56
5 DESCRIPTION OF PERSONNEL AND FACILITIES ..... 65
5.1 Supervisor ..... 65
5.2 Co-Supervisor ..... 65
6 REFERENCES ..... 66
7 APPENDICES ..... 69

# LIST OF FIGURES

Figure 3.1: System Architecture

Figure 3.2: System Workflow

Figure 3.3: AEIF Architecture

Figure 3.4: AED Architecture

Figure 3.5: SCP Decision Flow

Figure 3.6: Overall Implementation Architecture of the Proposed System

Figure 3.7: User Dashboard Interface

Figure 3.8: Food Registration Interface

Figure 3.9: Expiry Prediction Output Interface

Figure 3.10: Inventory Management Interface

Figure 3.11: Feedback-Based Personalization Interface

Figure 3.12: Personalized Expiry Prediction Output Interface

Figure 4.1: Expiry Distribution by Storage Type

Figure 4.2: Expiry Distribution by Category

Figure 4.3: Temperature vs Expiry

Figure 4.4: Humidity vs Expiry

Figure 4.5: Predicted Expiry vs Base Expiry

Figure 4.6: Distribution of Expiry Ratio

# LIST OF TABLES

Table 1: Comparative Analysis of Existing Food Expiry Prediction Approaches and Identified Research Gaps……. 09

Table 2: System Testing……48

Table 3: Model Comparison……..56

# LIST OF APPENDICES

Appendix 1 -- Plagiarism Report ..... 12

# LIST OF ABBREVIATIONS

**AI** – Artificial Intelligence

**ML** – Machine Learning

**NLP** – Natural Language Processing

**IoT** – Internet of Things

**FAO** – Food and Agriculture Organization

**API** – Application Programming Interface

**AEIF -**Adaptive Expiry Intelligence Framework

**AED** – Adaptive Expiry Adjustment

**SCP** – Smart Consumption Priority

# 1. INTRODUCTION

Food waste has emerged as one of the most persistent global sustainability challenges, affecting food security, environmental stability, and household economics at the same time. Recent international reporting shows that households, food service providers, and retailers continue to discard massive quantities of edible food, with household waste representing a particularly significant share of the problem. The consequences of this pattern extend beyond the direct loss of food, since wasted food also embodies wasted water, land, labor, energy, and logistics resources. At a time when hunger and food insecurity remain widespread, the continued disposal of usable food highlights the need for smarter and more context-aware food management systems. In practice, many households still rely on rough visual judgment, static package dates, or personal guesswork when deciding whether food should be consumed or discarded, which often leads either to premature disposal or unsafe consumption decisions [1], [2].

One of the major reasons for avoidable food waste is the mismatch between printed expiration labels and the actual condition of food under real storage environments. Conventional date labels are usually assigned under generalized assumptions about packaging, storage, and handling, but real household conditions are rarely constant. Temperature fluctuations, humidity variation, storage location, product type, and user handling behavior can all influence deterioration rates. As a result, two apparently similar food items may not remain safe or acceptable for the same duration. This limitation becomes even more significant in domestic settings, where refrigeration practices, pantry conditions, container usage, and repeated exposure to room temperature differ widely from one user to another. A static label therefore provides only limited decision support, especially when the goal is to estimate the remaining useful life of a food item after purchase and during everyday storage [3], [4].

The interpretation of date labels itself is also a well-documented source of confusion. Studies examining consumer understanding of date marking consistently report that many users do not clearly distinguish between quality-oriented labels and safety-oriented labels. In many situations, foods that are still acceptable for consumption are discarded because users interpret

“best before” or similar labels as strict indicators of danger rather than guidance on peak quality. This confusion creates a decision environment in which risk aversion encourages waste, especially for perishable products. From a system design perspective, this means that expiry support tools should not merely repeat package labels; instead, they should provide richer and more personalized reasoning that reflects both storage conditions and safety constraints. An intelligent expiry prediction system therefore has value not only as a technical forecasting tool but also as a practical decision aid that can reduce uncertainty for everyday users [3], [4].

In response to these challenges, the food research community has increasingly explored data-driven methods for quality monitoring and shelf-life prediction. Recent literature shows growing interest in applying artificial intelligence, machine learning, and hybrid analytical frameworks to estimate food quality changes, spoilage progression, and remaining shelf life under varying conditions. These approaches are particularly useful because food degradation is rarely governed by a simple linear relationship. Instead, it is influenced by interacting factors such as storage temperature, humidity, packaging conditions, product category, physicochemical characteristics, and microbial behavior. Reviews published in this area emphasize that machine learning techniques can improve predictive capability by learning nonlinear relationships directly from data, especially when traditional fixed-rule estimation becomes too rigid for diverse real-world scenarios [5], [6].

Several recent studies demonstrate the practical feasibility of predictive shelf-life modeling. For example, machine learning-based shelf-life estimation has been explored for fruits, vegetables, garlic, broccoli, marine fish, and meat products using combinations of environmental variables, imaging signals, physicochemical indicators, and quality measurements. These studies show that predictive models can capture spoilage trends more effectively than simplistic threshold-based methods when sufficient contextual information is available. At the same time, the literature also reveals an important limitation: most existing works focus on specific product groups, laboratory-grade sensing environments, or industrial quality-control settings rather than user-centered household decision support. In other words, while many studies successfully model spoilage or shelf life, relatively few translate those

predictions into a personalized consumer-facing framework that can adapt over time to individual storage behavior and feedback [6], [8], [9], [10].

A further limitation in many existing food shelf-life prediction studies is the absence of a robust safety-oriented decision layer. Purely data-driven predictions, although useful, may occasionally generate outputs that are statistically plausible but operationally unsafe if they do not respect biological or domain-specific shelf-life constraints. For food applications, predictive performance alone is not sufficient; outputs must also remain bounded within realistic safety ranges. This is particularly important in domestic systems, where end users may interpret model outputs as trustworthy guidance for consumption. Therefore, an effective food expiry system should combine predictive intelligence with domain rules that prevent unsafe overestimation while still avoiding unnecessary conservatism that increases waste. This need creates a strong case for hybrid intelligent architectures that integrate machine learning with explicit safety logic [5], [6].

From a modeling perspective, structured food expiry datasets also present specific technical requirements. Such datasets commonly contain mixed feature types, including categorical descriptors such as food item and storage class, along with numerical variables such as temperature, humidity, or quantity. In this setting, gradient boosting methods have shown strong performance because they can model nonlinear interactions effectively while remaining suitable for tabular data. CatBoost is especially relevant because it was designed to improve boosting quality on datasets containing categorical features and to reduce prediction shift through ordered boosting. These properties make it well suited to expiry prediction problems where food identity, category, and storage representation are important explanatory signals and where interactions among features are unlikely to be purely linear [7].

Motivated by these gaps, this study presents a hybrid intelligent system for personalized food expiry prediction. The proposed approach is built around an AEIF that combines three complementary capabilities: machine learning-based shelf-life prediction, rule-based biological safety validation, and adaptive user-driven personalization. Within this framework, a CatBoost regression model is used to estimate the remaining shelf life of food items using structured food and environmental inputs. The prediction is then constrained through safety

rules to ensure that the final output remains biologically and practically acceptable. Beyond this, the system introduces an AED mechanism that learns from user feedback over time, thereby enabling personalized refinement rather than one-time static estimation. A SCP layer is also incorporated to translate predicted shelf life into actionable urgency signals that can help users consume items in a timely manner and reduce avoidable disposal. Together, these components move the system beyond a standalone predictor and position it as a practical household decision-support tool for safer and more efficient food management.

The significance of this work lies in its attempt to bridge the gap between predictive modeling and real-world usability. Rather than treating expiry estimation as a purely technical regression problem, the proposed system recognizes that usable household intelligence must be accurate, safe, adaptive, and interpretable enough to support everyday action. By combining data-driven modeling with safety constraints and personalized feedback adjustment, the research contributes toward a more realistic and application-oriented approach to food shelf-life management. In doing so, it addresses a problem space that is highly relevant to sustainability, consumer decision-making, and the broader goal of reducing food waste through intelligent digital systems.

## 1.1 Background & Literature Survey

Understanding food spoilage and shelf-life estimation requires a multidisciplinary perspective that combines food science, environmental dynamics, and computational modeling. Food spoilage is a natural and inevitable process driven by biological, chemical, and physical changes that progressively degrade food quality, safety, and nutritional value. Microbial activity, enzymatic reactions, and oxidation processes play a central role in this deterioration, leading to changes in taste, texture, color, and overall edibility. Environmental conditions such as temperature, humidity, oxygen exposure, and light significantly accelerate these processes, making shelf-life highly context-dependent rather than fixed. For instance, increased temperature can enhance microbial growth rates, while high humidity promotes mold formation, both of which directly reduce the safe consumption period of food products [10].

This variability highlights a fundamental limitation in traditional food management practices, where static expiry labels are used despite dynamic real-world storage conditions.

Conventional approaches to shelf-life estimation have historically relied on laboratory-based testing methods such as microbiological assays, chemical stability analysis, and sensory evaluation. These techniques provide scientifically validated results under controlled conditions and are often supported by kinetic modeling approaches, including Arrhenius-based reaction rate models. While such methods offer strong theoretical foundations, they are inherently limited in their ability to adapt to fluctuating environmental conditions encountered during transportation, storage, and household usage. As noted in recent studies, these models often assume ideal and constant storage environments, which rarely reflect real-life scenarios, leading to either overly conservative or potentially unsafe estimations [11]. Furthermore, these traditional methods are time-consuming, expensive, and often destructive, making them unsuitable for continuous monitoring or large-scale real-time applications.

To overcome these limitations, sensor-based monitoring systems have been introduced as an intermediate solution for improving food quality tracking. Technologies such as IoT-enabled temperature and humidity sensors, gas detection systems, and RFID-based monitoring allow continuous observation of environmental conditions throughout the food lifecycle. These systems improve traceability and enable early detection of deviations from recommended storage conditions. However, their functionality is typically limited to threshold-based alerts rather than predictive estimation. In other words, while they can indicate when conditions become unsafe, they do not inherently provide an estimate of how long a product will remain consumable. This restricts their usefulness in proactive decision-making scenarios such as inventory prioritization or waste reduction strategies [12].

The emergence of artificial intelligence and machine learning has significantly transformed the landscape of food spoilage prediction and shelf-life estimation. AI-driven approaches leverage large volumes of heterogeneous data, including environmental conditions, microbial behavior, and historical spoilage patterns, to model complex, non-linear relationships that cannot be captured using traditional methods. Machine learning models such as regression algorithms, decision trees, random forests, and neural networks have been widely applied to

predict food quality attributes and remaining shelf life. These models are particularly effective because they can learn directly from data, adapt to varying conditions, and continuously improve as more data becomes available. Studies have shown that AI-based systems can provide dynamic shelf-life predictions that adjust according to real-time storage conditions, thereby reducing food waste and improving safety outcomes [13].

Recent advancements further extend these capabilities through deep learning and hybrid AI models. CNNs have been used to detect visual signs of spoilage such as discoloration and mold growth, while RNNs are applied to analyze time-series environmental data for predicting spoilage progression. Hybrid approaches combine multiple data sources, including sensor data, image analysis, and historical records, to improve prediction accuracy and robustness. These models have demonstrated strong potential in real-world applications such as smart packaging, automated food inspection, and supply chain optimization. For example, AI-integrated smart packaging systems can monitor temperature, humidity, and gas emissions in real time and provide dynamic freshness updates to consumers and stakeholders [14].

Despite these advancements, several limitations remain in current machine learning-based shelf-life prediction approaches. Many existing studies focus on specific food categories, limiting their generalizability across diverse products. Additionally, the quality and availability of training data significantly affect model performance, as food spoilage behavior varies widely depending on composition, storage conditions, and regional factors. Models trained under specific environmental conditions may not perform accurately in different climates or usage contexts. Another critical challenge is the lack of interpretability in complex models, particularly deep learning systems, which are often considered “black boxes.” This lack of transparency can reduce user trust and hinder adoption in safety-critical applications such as food consumption decisions [15].

A key advancement in recent literature is the incorporation of environmental exposure modeling and time-dependent analysis. Since food degradation is inherently a temporal process, several studies have introduced cumulative exposure indices and time-temperature integrals to better represent the effect of environmental stress over time. For instance, the study “Machine Learning-Enhanced Shelf-Life Prediction Model for Packaged Foods Using

Environmental Cues" demonstrates how combining temperature and humidity histories into an exposure index can significantly improve prediction accuracy by capturing cumulative degradation effects rather than relying on instantaneous measurements [16]. Additionally, the architecture presented in this work highlights a complete pipeline integrating environmental sensors, preprocessing modules, machine learning models, and decision-support outputs, emphasizing the importance of end-to-end system design for practical deployment .

Another important direction in the literature is the integration of AI with IoT and smart food systems. AI-powered spoilage prediction models are increasingly embedded into supply chain operations, retail systems, and even household appliances. These systems enable real-time monitoring, predictive analytics, and automated decision-making, contributing to improved food safety and reduced waste. For example, AI-driven refrigerators and mobile applications can track stored food items, predict expiry dates dynamically, and recommend consumption strategies to minimize waste. Similarly, AI-based supply chain systems can optimize inventory management by prioritizing items based on predicted spoilage risk, thereby enhancing efficiency and sustainability [17].

However, despite the rapid progress in AI-based shelf-life prediction, a significant research gap remains in developing systems that are both highly accurate and practically usable in real-world consumer environments. Many existing models focus primarily on prediction accuracy without incorporating safety constraints, personalization, or interpretability. In real-world applications, especially at the household level, users require systems that not only predict shelf life but also ensure safety, adapt to individual usage patterns, and provide actionable insights. This gap highlights the need for hybrid intelligent frameworks that integrate machine learning predictions with domain knowledge, safety rules, and user feedback mechanisms [18].

In summary, the literature indicates a clear evolution from static, laboratory-based shelf-life estimation methods toward dynamic, data-driven, and intelligent systems. While traditional methods provide foundational scientific understanding, they lack adaptability to real-world conditions. Sensor-based systems improve monitoring but fall short in predictive capability. Machine learning and AI approaches address many of these limitations by enabling adaptive, data-driven predictions, yet challenges related to generalization, interpretability, and practical

usability persist. These limitations motivate the development of advanced hybrid frameworks, such as the proposed system in this study, which aims to combine predictive intelligence with safety validation and personalized adaptation to deliver a more reliable and user-centric solution for food expiry prediction.

## 1.2 Research Gap

Despite significant advancements in food shelf-life estimation through traditional methods, sensor-based monitoring, and machine learning-driven approaches, several critical limitations persist in existing systems. Most current solutions focus primarily on predictive accuracy using either static rule-based models or data-driven machine learning techniques, but they often fail to integrate both effectively within a unified framework. Traditional shelf-life estimation methods rely on controlled laboratory conditions and generalized assumptions, which do not reflect the variability of real-world storage environments. While machine learning models have improved prediction capabilities by capturing nonlinear relationships among environmental and product-specific factors, many existing studies are limited to specific food categories and lack generalizability across diverse household scenarios. Furthermore, these models are typically developed as standalone predictors without incorporating domain-specific safety constraints, which raises concerns about the reliability of predictions in safety-critical contexts such as food consumption decisions.

Another major gap lies in the absence of personalization in existing food expiry prediction systems. Most current approaches generate uniform predictions that do not adapt to individual user behavior, storage habits, or consumption patterns. In real-world settings, food degradation is influenced not only by environmental factors but also by user-specific practices such as frequency of refrigeration, handling conditions, and storage consistency. However, existing systems rarely incorporate feedback mechanisms that allow models to learn and evolve based on user interactions. This lack of adaptability limits the practical usefulness of these systems, as they cannot provide context-aware predictions tailored to individual households. Additionally, while some studies incorporate environmental monitoring through IoT-based sensors, they typically focus on real-time condition tracking rather than long-term predictive adaptation or user-centric learning.

Moreover, existing literature largely overlooks the importance of integrating decision-support mechanisms that translate predictive outputs into actionable insights. Even when accurate shelf-life predictions are available, users are often left without guidance on how to prioritize consumption or manage their food inventory effectively. The absence of prioritization frameworks results in missed opportunities to reduce food waste through informed decision-making. Additionally, many machine learning models used in this domain suffer from limited interpretability, reducing user trust and hindering adoption in practical applications. The lack of explainable and transparent systems further emphasizes the need for structured frameworks that balance predictive performance with usability and safety.

Therefore, there is a clear need for a comprehensive and hybrid intelligent system that not only predicts food expiry accurately but also ensures biological safety, adapts to individual user behavior, and provides actionable consumption guidance. Addressing these gaps requires the integration of machine learning models with rule-based safety validation and adaptive feedback mechanisms within a unified architecture. This research aims to bridge these limitations by proposing an Adaptive Expiry Intelligence Framework that combines predictive modeling, safety constraints, personalization, and consumption prioritization into a cohesive and practical solution for real-world food management.

Table 1: Comparative Analysis of Existing Food Expiry Prediction Approaches and Identified Research Gaps

| System / Study | Approach Used | Key Strength | Limitation | Gap Identified | Your System Contribution |
|---|---|---|---|---|---|
| **Baranyi & Roberts (1994)** [12] | Kinetic microbial growth modeling | Scientifically validated shelf-life estimation | Assumes constant conditions | No adaptability to real-world environments | ML-based dynamic prediction (AEIF) |
| **Ruby Elizabeth et al. (2026)** [11] | ML with environmental cues | Uses temperature & humidity for prediction | No safety validation layer | Predictions may exceed safe limits | Safety-bounded prediction (AEIF) |

| | | | | | |
|---|---|---|---|---|---|
| **Shehzad (2025)** [10] | AI-based spoilage prediction | Captures nonlinear spoilage patterns | Lacks personalization & feedback learning | No user-specific adaptation | AED personalization layer |
| **Chen & Guestrin (XGBoost, 2016)** [14] | Gradient boosting ML model | High predictive performance | Pure ML, no domain constraints | No biological safety or real-world usability | Hybrid ML + rule-based system |
| **Zhang et al. (2026)** [16] | ML for food aging classification | Effective classification of spoilage | Category-specific, limited generalization | Not scalable across all food types | Generalized multi-food prediction |
| **Nychas et al. (2021)** [17] | Data-driven food analysis systems | Integrates food data analytics | No decision support mechanisms | No consumption prioritization | SCP (Smart Consumption Priority) |
| **IoT Smart Food Monitoring Systems** [18] | Sensor-based tracking (temp, humidity) | Real-time monitoring | No predictive intelligence | Cannot estimate remaining shelf life | Predictive + adaptive framework |
| **Deep Learning-Based Spoilage Detection (CNN/RNN)** [13] | Image + time-series modeling | Detects visual spoilage patterns | Black-box, low interpretability | No user trust, no explainability | Structured hybrid + interpretable outputs |
| **Existing Expiry Label Systems** | Static rule-based labeling | Simple and widely used | Fixed, inaccurate in real conditions | No adaptability, no intelligence | Fully dynamic personalized prediction |

### 1.3 Research Problem

The increasing volume of food waste at the household level highlights a critical need for more accurate and context-aware methods of estimating food shelf life. Despite the availability of printed expiry labels and traditional shelf-life estimation techniques, these approaches remain insufficient for real-world decision-making due to their reliance on generalized assumptions and static conditions. Conventional models, including laboratory-based microbial growth

models, assume controlled storage environments and fail to account for variations in temperature, humidity, and handling practices that occur in everyday usage [10], [12]. As a result, these methods often produce estimations that are either overly conservative, leading to premature disposal of edible food, or potentially unsafe, increasing the risk of consumption beyond safe limits. This mismatch between theoretical estimation and practical conditions creates uncertainty for users and contributes significantly to avoidable food waste.

Although machine learning approaches have improved the ability to model complex relationships between environmental factors and food degradation, existing systems still operate primarily as standalone predictive tools without addressing key practical requirements. Most models generate generalized outputs that do not adapt to individual user behavior, storage habits, or contextual variations, limiting their effectiveness in household environments [13], [16]. Furthermore, these systems rarely incorporate biological safety constraints, meaning that predictions, while statistically accurate, may not always align with safe consumption boundaries. The absence of adaptive learning mechanisms also prevents continuous improvement of predictions based on real user feedback, reducing long-term reliability. Additionally, current solutions lack structured decision-support capabilities that can guide users in prioritizing food consumption based on urgency, leaving the final decision to subjective judgment [17], [18].

Therefore, the core research problem addressed in this study is the development of a reliable, adaptive, and safe food expiry prediction system that can operate effectively under real-world conditions. Specifically, the challenge lies in designing a system that not only predicts the remaining shelf life of food items using environmental and contextual data but also ensures that predictions remain within biologically safe limits, dynamically adapts to individual user behavior, and provides actionable insights to support timely consumption decisions. Addressing this problem requires the integration of machine learning techniques with rule-based safety validation and user-driven feedback mechanisms within a unified hybrid framework. The inability of existing systems to simultaneously achieve prediction accuracy, safety assurance, personalization, and decision support underscores the necessity for the proposed approach in this research.

# 2. OBJECTIVES

## 2.1 Main Objectives

The primary objective of this research is to design and develop a hybrid intelligent system for personalized food expiry prediction that accurately estimates the remaining shelf life of food items under real-world storage conditions while ensuring safety and practical usability. The proposed system aims to integrate machine learning-based prediction with domain-specific safety validation and adaptive user-driven learning to overcome the limitations of traditional and standalone predictive approaches. In addition to improving prediction accuracy, the system is intended to provide context-aware and user-specific expiry estimations that reflect environmental variations such as temperature and humidity, as well as individual storage and handling behaviors. By combining predictive intelligence with safety constraints and decision-support mechanisms, the research seeks to deliver a comprehensive solution that not only minimizes food waste but also enhances user confidence in food consumption decisions within household environments.

## 2.2 Specific Objectives

- To develop a machine learning-based regression model capable of accurately predicting the remaining shelf life of food items using structured features such as food type, storage conditions, temperature, and humidity.

- To implement a biological safety validation mechanism that constrains predicted expiry values within realistic and safe limits, ensuring that outputs remain aligned with acceptable consumption boundaries.

- To design and integrate an adaptive personalization mechanism that continuously refines expiry predictions based on user feedback and historical interaction patterns, enabling user-specific prediction adjustments.

• To develop a decision-support component that prioritizes food items based on urgency, allowing users to effectively manage food consumption and reduce waste through informed decision-making.

• To evaluate the performance of the proposed system by comparing it with multiple baseline models using appropriate evaluation metrics, in order to validate its accuracy, reliability, and practical applicability.

• To evaluate the system's performance in terms of accuracy, usability, and impact on food waste reduction, ensuring practical benefits for households.

# 3. METHODOLOGY

## 3.1 Methodology

### 3.1.1 System Overview

The proposed Personalized Food Expiry Prediction System is designed as a hybrid intelligent framework that integrates machine learning, rule-based safety validation, and adaptive user-driven personalization to provide accurate and context-aware expiry predictions.

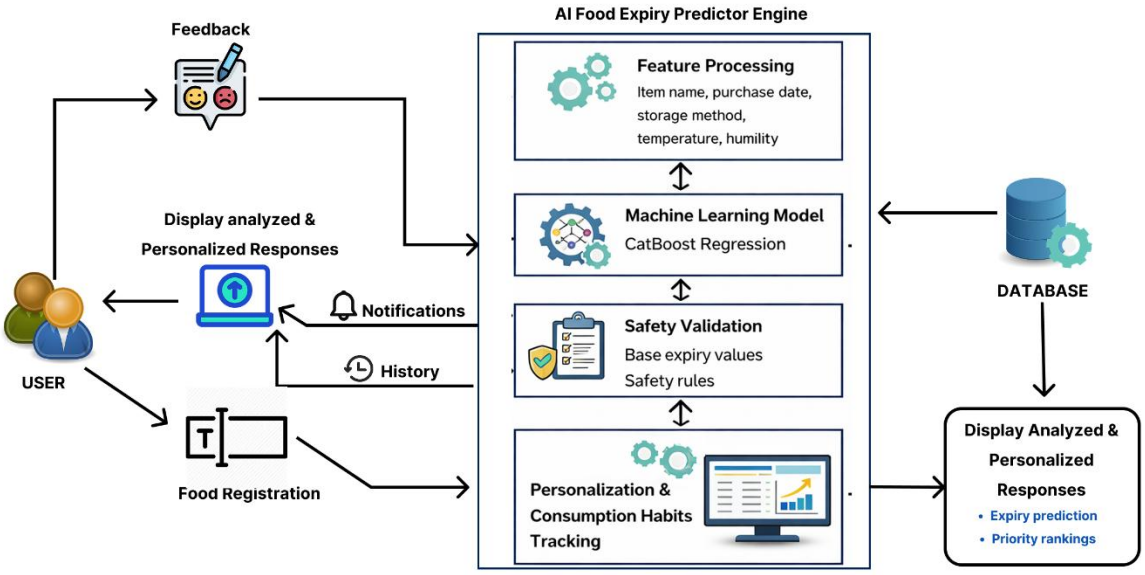


Figure 3.1: System Architecture

As illustrated in the system architecture diagram, the overall structure follows a modular pipeline in which user inputs are processed through multiple analytical layers before generating final outputs in the form of expiry predictions and consumption priorities. The system operates as a closed-loop intelligent environment, where user interaction, predictive modeling, and feedback continuously influence system performance and refinement.

The workflow begins with the user interacting with the system through the food registration interface, where essential information such as food item name, purchase date, storage method, and environmental conditions including temperature and humidity are provided. These inputs form the foundational data required for expiry prediction. Once the data is captured, it is passed to the feature processing module, where raw inputs are cleaned, standardized, and transformed into structured features suitable for machine learning. This stage ensures that categorical and numerical data are properly encoded and aligned with the trained model's feature space, enabling consistent and reliable predictions.

Following feature processing, the structured data is forwarded to the core machine learning component, which utilizes a CatBoost regression model to estimate the remaining shelf life of the food item. The model analyzes complex interactions between input variables, including storage conditions and environmental factors, to generate an initial prediction of expiry duration. However, unlike conventional systems that rely solely on model outputs, the proposed framework incorporates an additional safety validation layer. This layer enforces biologically grounded constraints using predefined base expiry values and safety rules, ensuring that the predicted shelf life remains within realistic and safe consumption limits. By applying these constraints, the system prevents overestimation and enhances reliability in practical usage.

In parallel, the system integrates a personalization and consumption habits tracking module, which continuously monitors user interactions, feedback, and historical consumption patterns. This module enables adaptive learning by adjusting future predictions based on user-specific behavior, thereby improving long-term accuracy and relevance. Feedback provided by users, such as early spoilage or extended usability, is fed back into the system, creating a feedback

loop that refines prediction accuracy over time. This adaptive mechanism transforms the system from a static predictor into a dynamic and evolving intelligent assistant.

The processed results are then stored and managed within the database, which maintains records of food items, prediction history, user feedback, and system outputs. This data persistence layer supports functionalities such as history tracking, notifications, and continuous model interaction. The final outputs, including predicted expiry dates and priority rankings, are delivered to the user through the display interface. Additionally, the system provides notifications and alerts to inform users about approaching expiry deadlines, enabling timely consumption and reducing food waste.

Overall, the system architecture demonstrates a seamless integration of input acquisition, intelligent processing, safety validation, personalization, and output delivery within a unified framework. By combining machine learning predictions with rule-based logic and user-centric adaptation, the system not only enhances prediction accuracy but also improves usability and decision-making support. This holistic design ensures that the system effectively addresses real-world challenges associated with food expiry management while maintaining safety, adaptability, and practical relevance.

### 3.1.2 System Workflow

The system workflow of the proposed Personalized Food Expiry Prediction System is designed to ensure a structured, adaptive, and user-centric process that transforms raw user inputs into accurate, safe, and actionable expiry predictions. The workflow follows a sequential yet interconnected pipeline, beginning with user interaction and progressing through multiple intelligent processing layers before generating final outputs.

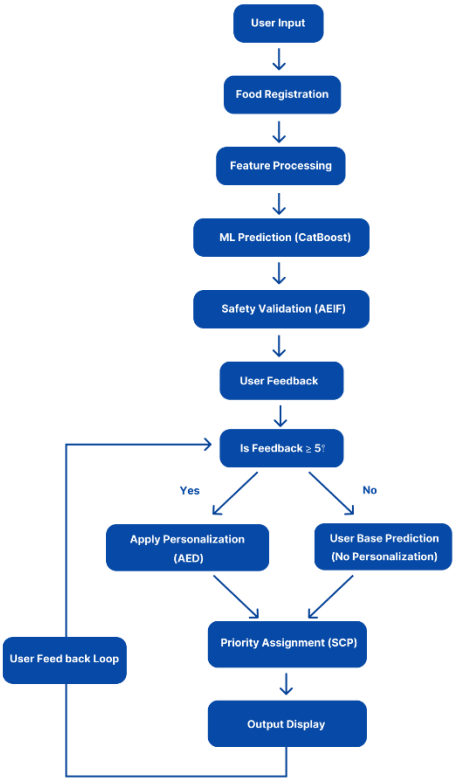


Figure 3.2: System Workflow

As illustrated in Fig. 3.2, the system operates as a continuous feedback-driven loop in which user inputs, predictive modeling, safety validation, and personalization mechanisms interact dynamically to improve prediction accuracy and usability over time.

The workflow begins at the food registration stage, where the user provides essential information related to stored food items. This includes attributes such as the food item name, purchase date, storage type (e.g., pantry, refrigerator, or freezer), and environmental parameters such as temperature and humidity. These inputs form the foundational dataset required for prediction and reflect real-world storage conditions, which are known to significantly influence food degradation rates. Unlike traditional systems that rely solely on static expiry labels, the proposed system captures contextual and dynamic inputs to enable more accurate estimation of remaining shelf life.

Once the input data is collected, it is passed to the feature processing stage, where raw inputs are transformed into structured features suitable for machine learning. This step involves data cleaning, normalization, and encoding of categorical variables, ensuring consistency with the

trained model's feature space. Feature engineering techniques such as one-hot encoding and base expiry mapping are applied to enhance the representation of food characteristics and storage conditions. Proper feature processing is essential for enabling machine learning models to effectively capture nonlinear relationships among variables, as demonstrated in prior studies on food spoilage prediction.

Following feature processing, the structured data is forwarded to the machine learning prediction module, which forms the core analytical component of the system. The proposed system utilizes a CatBoost regression model to estimate the number of days remaining until food expiry. CatBoost is selected due to its ability to handle categorical features efficiently and model complex interactions within structured datasets. The model processes the input features and generates an initial prediction of shelf life, which represents the raw estimation before any domain-specific constraints are applied. This data-driven approach allows the system to adapt to varying environmental conditions and provide more accurate predictions compared to static or rule-based methods.

However, to ensure that predictions remain biologically safe and practically reliable, the workflow incorporates a safety validation stage as part of the AEIF. In this stage, the raw model output is adjusted using predefined base expiry values and safety rules derived from food science principles. These constraints ensure that predicted expiry values do not exceed safe consumption limits or fall below realistic thresholds. By combining machine learning predictions with rule-based validation, the system addresses a key limitation identified in existing literature, where purely data-driven models may produce unsafe or unrealistic outputs.

In parallel, the system integrates a personalization mechanism through AED module. This component continuously learns from user feedback and historical interaction data to refine future predictions. For instance, if a user indicates that a food item spoiled earlier or lasted longer than predicted, the system adjusts its internal parameters accordingly. This adaptive learning process enables the model to align with individual user behavior, addressing the lack of personalization observed in existing systems. Over time, this feedback-driven approach

enhances prediction accuracy and relevance for each user, transforming the system into a dynamic and evolving intelligent assistant.

To ensure stability and prevent premature or unreliable adjustments, the personalization mechanism is activated only after a sufficient amount of user feedback has been collected. Specifically, the system requires a minimum threshold of feedback instances (e.g., five or more observations) before applying item-level adaptive adjustments. If the available feedback count is below this threshold, the system relies solely on the baseline prediction generated by the AEIF without applying personalization. This design choice ensures that the adjustment process is supported by sufficient data, thereby avoiding overfitting and maintaining prediction reliability during the early stages of system usage. Once the threshold is met, the AED module dynamically refines predictions based on accumulated feedback, enabling accurate and stable personalization over time.

After safety validation and personalization adjustments, the system proceeds to the decision-support stage, implemented through the SCP module. In this stage, the predicted expiry values are translated into actionable insights by assigning priority levels to food items based on urgency. Items nearing expiry are marked with higher priority, enabling users to make informed consumption decisions and reduce food waste. The integration of such prioritization mechanisms is essential for improving real-world usability, as highlighted in prior research on intelligent food management systems.

Finally, the processed results are delivered to the user through the output interface, which displays predicted expiry dates, remaining days, and priority rankings. Additional functionalities such as notifications and alerts are triggered to inform users about approaching expiry deadlines. All data, including predictions, user inputs, and feedback, are stored in the system database, enabling history tracking and continuous model interaction. The workflow concludes with a feedback loop, where user responses are fed back into the system to further refine predictions, ensuring continuous improvement and adaptation.

Overall, the system workflow demonstrates a seamless integration of data acquisition, intelligent processing, safety validation, personalization, and decision support within a unified

hybrid framework. By combining machine learning with domain knowledge and user-driven adaptation, the proposed workflow effectively addresses the limitations of existing approaches and provides a comprehensive solution for accurate and practical food expiry prediction.

### 3.1.3 Data Featuring

The effectiveness of the proposed food expiry prediction system is fundamentally dependent on the quality and relevance of the data used for model training. In this study, a structured tabular dataset was utilized to capture the relationships between food characteristics, storage conditions, and shelf-life duration. The primary dataset was obtained from a publicly available food expiry prediction dataset, which contains real-world observations of food items stored under varying environmental conditions. After preprocessing and removal of inconsistent or duplicate records, the final dataset consisted of approximately 2048 samples, ensuring sufficient data diversity for model training and evaluation.

The dataset includes a combination of categorical and numerical features that collectively describe the state and storage environment of food items. Key features include the food item name, category, and storage type, which provide contextual information about the product. In addition, environmental attributes such as temperature and humidity are incorporated, as these factors are known to significantly influence food degradation rates by affecting microbial activity and chemical reactions. Additional attributes such as quantity and storage conditions further enhance the model's ability to capture real-world variability. To support safety validation within the system, a secondary dataset containing base expiry values for different food items under standard storage conditions was also integrated. This dataset provides reference expiry durations for pantry, refrigeration, and freezer environments, enabling the system to apply biologically grounded constraints during prediction.

The target variable used in this study is the number of days until expiry, represented as a continuous numerical value. This variable reflects the remaining shelf life of each food item under specific conditions and serves as the dependent variable for the regression model. By formulating the problem as a regression task, the system is able to generate precise and continuous predictions rather than discrete classifications, which improves the practical

usability of the output. The selection of relevant features and an appropriate target variable ensures that the dataset effectively supports the learning of complex relationships between environmental factors and food spoilage, as highlighted in prior research on machine learning-based shelf-life prediction.

### 3.1.4 Data preprocessing

Data preprocessing is a critical step in ensuring the quality, consistency, and reliability of the dataset before it is used for model training. Raw data collected from multiple sources often contains inconsistencies, missing values, and redundant information, which can negatively impact model performance if not properly addressed. Therefore, a systematic preprocessing pipeline was implemented to clean, transform, and standardize the dataset in preparation for machine learning.

The initial step involved data cleaning, where duplicate records and inconsistent entries were identified and removed. This process ensures that the dataset does not contain redundant samples that could bias the model during training. Additionally, irrelevant attributes such as transaction identifiers, textual notes, or unused metadata fields were excluded from the feature set to reduce noise and improve computational efficiency. Standardization of data formats, particularly for categorical variables such as food names and storage types, was also performed to maintain consistency across all records.

Handling missing values was another essential component of the preprocessing stage. In cases where the target variable was missing, the corresponding records were removed to preserve the integrity of the regression task. For input features with missing values, appropriate imputation techniques were applied. Numerical features such as temperature and humidity were filled using default or statistically representative values, while categorical features were handled through encoding strategies that ensured no loss of structural information. This approach maintains dataset completeness while minimizing distortion in feature distribution, which is important for accurate model learning.

To enable effective processing by the machine learning model, categorical variables were transformed into numerical representations through encoding techniques. One-hot encoding

was applied to features such as food item names and categories, converting them into binary vectors that allow the model to distinguish between different classes. Storage types were similarly encoded to represent different environmental conditions. This transformation is particularly important for models dealing with structured tabular data, as it allows the algorithm to interpret categorical relationships without imposing unintended ordinal assumptions. The use of proper encoding techniques has been shown to significantly improve model performance in regression-based prediction tasks.

Overall, the preprocessing pipeline ensures that the dataset is clean, complete, and structured in a manner suitable for machine learning. By addressing data quality issues and transforming features into meaningful representations, this stage plays a crucial role in enhancing the predictive capability and reliability of the proposed system.

### 3.1.5 Feature Engineering

Feature engineering plays a critical role in enhancing the predictive capability of the proposed food expiry prediction system by transforming raw input data into meaningful representations that can be effectively utilized by the machine learning model. Given the structured and heterogeneous nature of the dataset, which consists of both categorical and numerical attributes, a systematic feature engineering process was implemented to capture the complex relationships between food characteristics, environmental conditions, and shelf-life duration.

One of the primary steps in feature engineering involved the transformation of categorical variables into numerical representations. Features such as food item names and categories were converted using one-hot encoding, resulting in a high-dimensional feature space where each unique item or category is represented as a binary vector. This approach allows the model to differentiate between distinct food types without introducing unintended ordinal relationships. Similarly, storage conditions, including pantry, refrigeration, and freezer environments, were encoded to reflect their impact on food preservation. The inclusion of these encoded features enables the model to learn how different storage environments influence the rate of food degradation, which is a key factor in shelf-life prediction.

In addition to categorical encoding, environmental attributes such as temperature and humidity were retained as continuous numerical features. These variables are critical for modeling food spoilage, as they directly influence microbial growth and chemical reactions within food products. By incorporating these features into the model, the system is able to capture nonlinear interactions between environmental conditions and food characteristics. Previous research has demonstrated that environmental factors significantly improve the accuracy of machine learning-based shelf-life prediction models by providing context-aware insights into spoilage dynamics.

A key contribution of the feature engineering process in this study is the incorporation of a derived feature representing the base expiry duration of food items. This feature is generated by mapping each food item to its corresponding standard shelf-life value under ideal storage conditions using an external base expiry dataset. The base expiry values differ depending on the storage type, such as pantry, refrigerator, or freezer, and serve as a reference point for both prediction and safety validation. By integrating this feature into the model, the system gains additional domain knowledge that helps guide predictions toward realistic and biologically plausible values. This approach bridges the gap between purely data-driven modeling and domain-informed estimation, improving both accuracy and reliability.

Furthermore, feature normalization and scaling were considered to ensure that numerical variables are within comparable ranges, thereby preventing dominance of features with larger magnitudes. Although tree-based models such as CatBoost are generally robust to feature scaling, maintaining consistent feature distributions contributes to stable model behavior and improves interpretability. Missing or undefined feature values were also handled during this stage by applying appropriate default values or statistical imputation, ensuring that all input vectors remain complete and consistent.

The final feature set used for model training consists of a high-dimensional representation combining encoded categorical variables, continuous environmental attributes, and derived domain-specific features. In total, the engineered dataset includes over 600 features, reflecting the diversity of food items and their associated characteristics. This rich feature space enables

the model to capture intricate interactions between variables, which is essential for accurate shelf-life prediction in complex real-world scenarios.

Overall, the feature engineering process enhances the representational power of the dataset by combining raw data transformation, environmental context integration, and domain knowledge incorporation. This comprehensive approach ensures that the machine learning model is provided with informative and relevant inputs, thereby improving prediction accuracy and supporting the effectiveness of the overall hybrid system.

### 3.1.6 Adaptive Expiry Intelligence Feature

#### 3.1.6.1 Conceptual Foundation and Motivation

The Adaptive Expiry Intelligence Framework is introduced as the core methodological contribution of this research, designed to address the fundamental limitations observed in existing food expiry prediction systems.

Current approaches typically fall into two categories:

- traditional rule-based systems
- purely data-driven machine learning models.

Rule-based systems rely on static assumptions derived from controlled laboratory conditions, which fail to capture the variability of real-world storage environments. Conversely, machine learning models, while capable of modeling complex nonlinear relationships, often lack domain-awareness and may generate predictions that are not aligned with biological safety constraints [5], [6].

This disconnect between predictive accuracy and practical safety necessitates the development of a hybrid framework that can effectively integrate both paradigms. The design of AEIF is therefore inspired by established principles in hybrid intelligent systems, where multiple reasoning mechanisms are combined to enhance robustness, adaptability, and

decision-making reliability [21]. Such hybridization is particularly critical in safety-sensitive domains, where decisions must be both data-driven and constrained by domain knowledge.

Furthermore, the incorporation of structured rule-based validation within a predictive framework aligns with foundational work in intelligent decision systems, where uncertainty and interpretability must be carefully balanced [22]. Recent advancements in machine learning also emphasize the importance of integrating constraints and prior knowledge to improve generalization and reduce prediction instability [25].

Therefore, AEIF is not derived from a single existing model but is instead formulated as a novel framework that synthesizes concepts from hybrid intelligence, safety-constrained learning, and adaptive decision-support systems. This integration enables the system to produce predictions that are not only statistically accurate but also biologically valid, interpretable, and practically usable.

#### 3.1.6.2 Layered Architecture and Operational Mechanism

The AEIF operates as a multi-layered processing architecture that transforms raw model predictions into validated and actionable outputs through sequential refinement stages. This structured design ensures that each stage contributes to improving prediction accuracy, safety, and reliability.

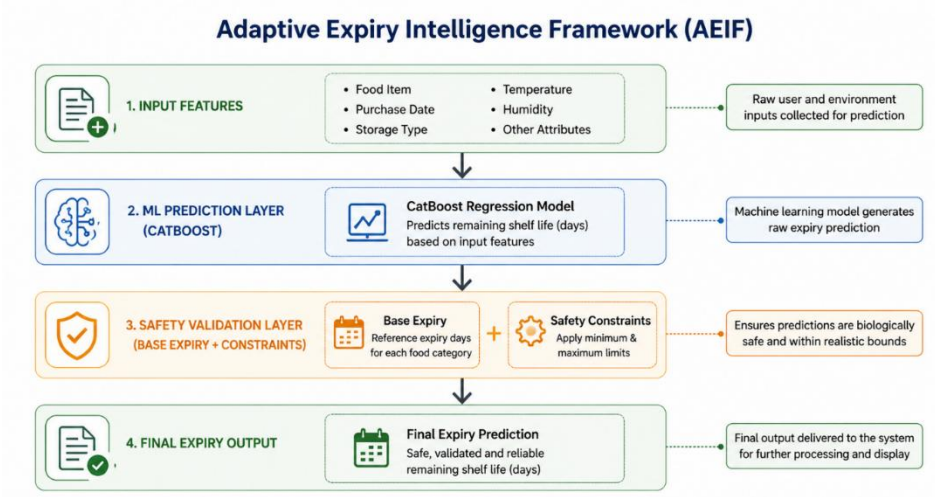


Figure 3.3: AEIF Architecture

The first stage of the framework is the Machine Learning Prediction Layer, where the CatBoost regression model generates an initial estimate of the remaining shelf life based on input features such as food type, storage conditions, temperature, and humidity. This layer captures complex nonlinear interactions within the data and provides a high-quality baseline prediction derived from statistical learning [13]. However, as widely recognized in machine learning literature, purely data-driven models may produce outputs that are statistically optimal but not necessarily safe or realistic in practical scenarios [17].

To address this limitation, the second stage introduces the Safety Validation Layer, which is the defining component of AEIF. In this stage, the raw prediction is evaluated against domain-specific constraints derived from base expiry values associated with each food item. These base values represent standard shelf-life durations under ideal storage conditions and serve as reference points for validation. The system enforces both lower and upper bounds to ensure that predictions remain within biologically acceptable limits. This constraint-based validation reflects established methodologies in intelligent decision systems, where rule-based reasoning is integrated to ensure safety and consistency [22].

The safety validation process is mathematically defined as:

$$\text{FinalExpiry} = \min(\max(\text{PredictedExpiry}, \alpha \times \text{BaseExpiry}), \beta \times \text{BaseExpiry})$$

where:

- $\alpha$ represents the minimum safety threshold (e.g., 0.6),
- $\beta$ represents the maximum allowable limit (e.g., 1.5),
- BaseExpiry denotes the standard reference shelf-life for the food item.

In addition to these constraints, manufacturer-provided expiry dates, when available, are applied as hard upper bounds, ensuring strict adherence to known safety limits. This multi-level validation mechanism effectively combines statistical learning with deterministic constraints, thereby improving both reliability and safety.

The final stage of AEIF is the Output Adjustment Layer, where the validated expiry value is finalized and forwarded to downstream modules such as personalization (AED) and prioritization (SCP). This layered architecture ensures that predictions are progressively refined rather than directly exposed, resulting in a robust and context-aware output.

#### 3.1.6.3 Technical Advantages and Research Contribution

The introduction of AEIF provides several significant technical advantages that distinguish the proposed system from existing approaches.

First, AEIF enhances prediction reliability by integrating machine learning outputs with domain-specific safety constraints. This hybrid approach reduces the risk of unsafe or unrealistic predictions, addressing a critical limitation of purely data-driven models. By enforcing biologically grounded boundaries, the framework ensures that all outputs remain within acceptable consumption limits.

Second, the framework improves robustness and stability. Machine learning models are inherently sensitive to data variability and noise; however, the inclusion of validation rules ensures that extreme or anomalous predictions are corrected before reaching the user. This contributes to consistent system performance across diverse environmental conditions.

Third, AEIF demonstrates strong alignment with hybrid learning principles, where combining multiple learning and reasoning strategies leads to improved generalization and adaptability [23]. The integration of structured constraints with predictive learning enhances the system's ability to handle uncertainty and real-world variability.

Fourth, the framework supports decision-oriented processing, enabling seamless integration with ranking and prioritization mechanisms such as SCP. This aligns with research in preference learning and decision systems, where outputs are not only predicted but also structured to support actionable decisions [24].

Fifth, AEIF improves interpretability and user trust. By clearly defining how predictions are generated, validated, and adjusted, the framework provides transparency that is often lacking

in black-box models. This is particularly important in food-related applications, where users rely on system outputs for safety-critical decisions.

Overall, AEIF represents a novel hybrid intelligence framework that bridges the gap between predictive modeling and domain-aware validation. By combining machine learning, safety constraints, and structured system logic, it provides a comprehensive and practical solution for food expiry prediction in real-world environments.

### 3.1.7 Adaptive Expiry Drift

#### 3.1.7.1 Concept and Motivation of AED

While the AEIF ensures that predictions are accurate and biologically safe, it does not inherently account for variability in individual user behavior and real-world usage patterns. In practical scenarios, food spoilage is influenced not only by environmental conditions but also by user-specific factors such as handling practices, storage consistency, frequency of exposure, and consumption habits. Traditional machine learning models, which are typically trained on generalized datasets, are unable to capture these personalized variations, resulting in predictions that may not fully align with actual user experiences.

To address this limitation, the AED module is introduced as a dynamic personalization layer that operates on top of the validated outputs generated by AEIF. The primary objective of AED is to enable the system to learn continuously from user feedback and adjust future predictions accordingly. This aligns with broader research in adaptive learning systems, where models evolve based on interaction data to improve long-term performance and user relevance [23]. By incorporating feedback-driven learning, AED transforms the system from a static predictive model into an adaptive intelligent assistant that evolves over time.

Furthermore, the integration of user feedback into predictive systems is consistent with principles in decision-support and preference learning, where user-specific patterns are leveraged to refine outputs and improve decision quality [24]. In the context of food expiry prediction, this means that the system can progressively align its predictions with the actual behavior and experiences of individual users, thereby increasing accuracy and usability.

#### 3.1.7.2 Operational Mechanism and Learning Process

The AED module operates as a feedback-driven adjustment layer that modifies the expiry predictions produced by AEIF based on historical user interactions. The process begins when a user provides feedback on a previously predicted food item. This feedback can indicate whether the item expired earlier than predicted, lasted longer than expected, or matched the predicted duration. Such feedback is used to compute the prediction error, which forms the basis for subsequent adjustments.

The prediction error is defined as the difference between the actual observed shelf life and the predicted shelf life:

$$\text{Error} = \text{ActualDays} - \text{PredictedDays}$$

Based on this error, AED updates an adjustment factor associated with either the specific food item or its broader category. To ensure stability and prevent abrupt changes in prediction behavior, the system employs a weighted update mechanism that combines previous adjustment values with new feedback information. The update rule is defined as:

$$\text{NewAdjustment} = \text{PreviousAdjustment} \times \lambda + (\text{Error} \times \eta)$$

where:

- $\lambda$ represents a retention factor (e.g., 0.95), preserving historical learning,
- $\eta$ represents an effective learning rate, controlling the influence of new feedback,
- Error captures the deviation between actual and predicted values.

This formulation ensures that the system gradually adapts to user behavior without overreacting to individual feedback instances. Additionally, the learning rate is dynamically adjusted based on the number of feedback samples available, reducing sensitivity as more data is accumulated. This reflects established principles in machine learning, where adaptive learning rates contribute to stable convergence and improved generalization [25], .

To further enhance reliability, AED employs a hierarchical adjustment strategy. When sufficient feedback is available for a specific food item, item-level adjustments are applied. Otherwise, category-level adjustments are used as a fallback mechanism. This ensures that the system can provide personalized predictions even when data is limited, while progressively refining accuracy as more feedback becomes available.

#### 3.1.7.3 Integration with AEIF and System Workflow

The AED module is tightly integrated with the AEIF pipeline, operating after the safety validation stage to ensure that all adjustments are applied within biologically safe limits. This design ensures that personalization does not override safety constraints, thereby maintaining the integrity of the system. The adjusted expiry value produced by AED is subsequently used for decision-support processes, including prioritization and notification.

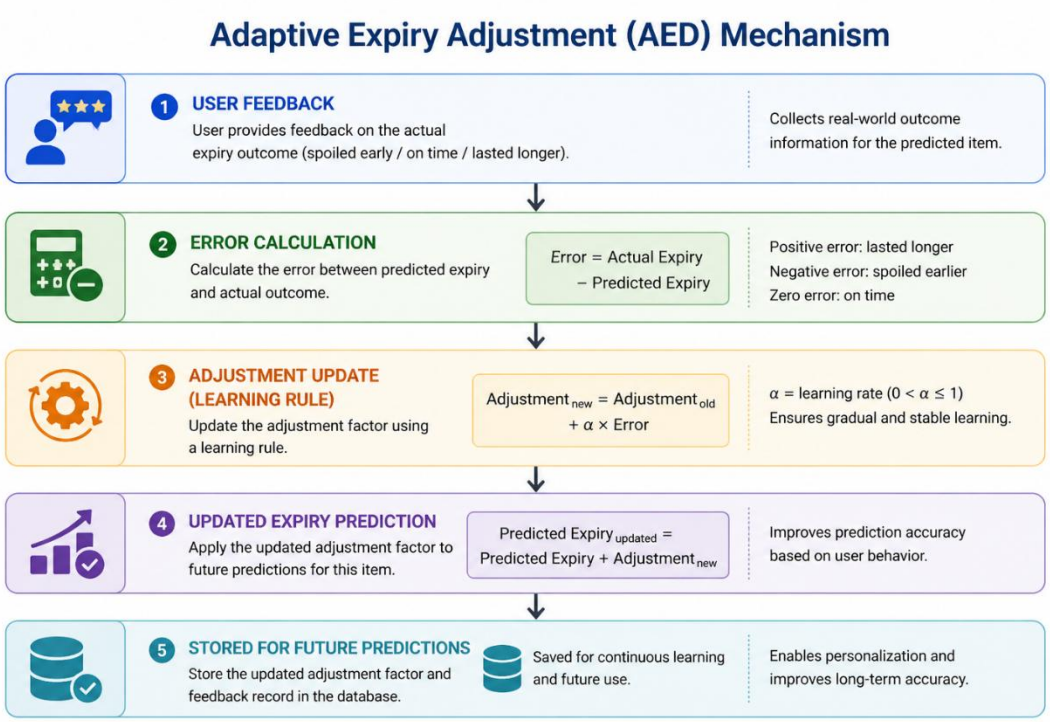


Figure 3.4: AED Architecture

From a system perspective, AED introduces a continuous feedback loop into the workflow. User interactions are not treated as isolated events but as valuable data points that contribute to ongoing system improvement. This aligns with modern intelligent system architectures,

where feedback loops are essential for achieving adaptability and long-term performance enhancement [21]. By integrating this loop, the system evolves over time, becoming increasingly aligned with real-world usage patterns.

#### 3.1.7.4 Technical Advantages and Contribution

The inclusion of AED provides several key advantages that significantly enhance the overall effectiveness of the proposed system. First, it enables true personalization, allowing the system to adapt predictions based on individual user behavior rather than relying solely on generalized data. This leads to more accurate and context-specific outputs.

Second, AED improves long-term prediction accuracy by continuously incorporating feedback into the learning process. As more data is collected, the system becomes increasingly refined, reducing prediction errors and improving reliability.

Third, the module enhances system adaptability, enabling it to function effectively across diverse users, environments, and usage patterns. This is particularly important in real-world applications, where variability is inherent and unavoidable.

Fourth, AED contributes to improved user engagement and trust. By allowing users to influence system behavior through feedback, the system becomes more interactive and responsive, encouraging continued usage and participation.

Finally, the integration of AED with AEIF demonstrates a complete hybrid intelligent framework that combines prediction, validation, and adaptation within a unified system. This multi-layered approach represents a significant advancement over existing models, which typically focus on only one aspect of the problem.

### 3.1.8 Smart Consumption Priority

#### 3.1.8.1 Concept and Motivation of SCP

While accurate and personalized expiry prediction is essential, it does not fully solve the problem of food waste unless users are provided with clear and actionable guidance on how

to utilize this information. In real-world scenarios, users often manage multiple food items simultaneously, making it difficult to determine which items should be consumed first. Even with accurate expiry dates, the absence of prioritization mechanisms forces users to rely on subjective judgment, which frequently leads to inefficient consumption patterns and increased food waste.

To address this limitation, the SCP module is introduced as a decision-support layer that transforms predicted expiry information into actionable insights. The primary objective of SCP is to assist users in managing their food inventory by assigning priority levels based on the urgency of consumption. This aligns with established principles in intelligent decision-support systems, where predictive outputs are enhanced through structured ranking and prioritization mechanisms to improve decision-making effectiveness [24].

Furthermore, integrating prioritization into predictive systems reflects broader trends in data-driven decision-making, where raw predictions are converted into meaningful actions. Research in machine learning emphasizes that the true value of predictive models lies not only in their accuracy but also in their ability to support real-world decisions [25]. In the context of food management, this translates to guiding users toward timely consumption, thereby reducing waste and improving resource utilization.

#### 3.1.8.2 Priority Scoring Mechanism

The SCP module operates by converting the final validated expiry predictions (after AEIF and AED processing) into a priority score that reflects the urgency of consumption. This is achieved by analyzing the number of remaining days before expiry and mapping it to predefined priority levels. The scoring mechanism is designed to be simple, interpretable, and effective for real-time decision-making.

The priority assignment is defined as follows:

- **High Priority (Critical Consumption Zone):**
  Food items with remaining shelf life less than or equal to 2 days are assigned the

highest priority. These items require immediate attention to prevent spoilage and waste.

- **Medium Priority (Moderate Consumption Zone):**
  Items with remaining shelf life between 2 and 5 days are assigned medium priority. These items should be scheduled for consumption in the near future.
- **Low Priority (Safe Zone):**
  Items with more than 5 days remaining are considered low priority, indicating that they are relatively stable and do not require immediate consumption.

This categorization can be mathematically expressed as:

$$\mathbf{SCP(x)} = \begin{cases} 1.0, & \text{if } x \leq 2 \\ 0.7, & \text{if } 2 < x \leq 5 \\ 0.4, & \text{if } x > 5 \end{cases}$$

where:

- X represents the number of days remaining before expiry,
- SCP(x) denotes the priority score assigned to the food item.

This scoring mechanism ensures that urgency is clearly quantified, enabling the system to rank food items effectively. The use of discrete priority levels simplifies user interpretation while maintaining sufficient granularity for decision-making.

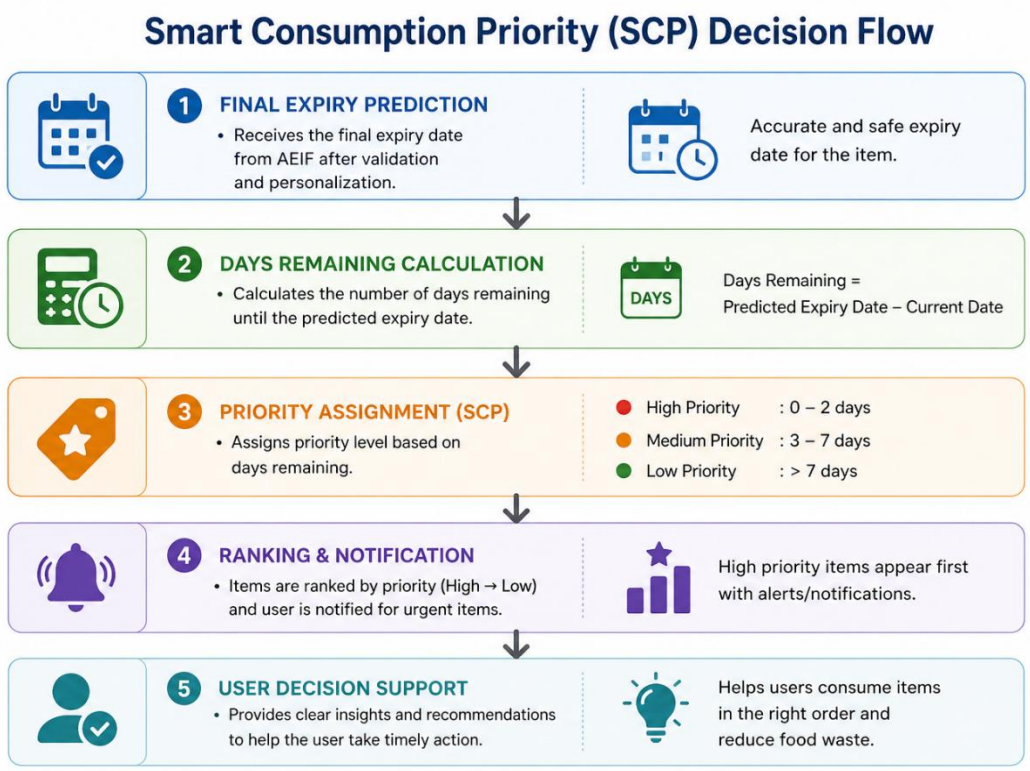


Figure 3.5: SCP Decision Flow

#### 3.1.8.3 Integration with System Workflow

The SCP module operates as the final decision-support layer within the overall system pipeline, receiving input from the AED module after all prediction and personalization adjustments have been applied. This ensures that priority assignments are based on the most accurate and user-adapted expiry values.

Once priority scores are computed, they are used to organize and present food items within the user interface. Items with higher priority are highlighted and positioned prominently, enabling users to quickly identify which items require immediate attention. Additionally, SCP supports notification mechanisms that alert users when items enter the high-priority zone, further enhancing proactive food management.

From a system perspective, SCP completes the transformation of raw data into actionable intelligence. While earlier stages of the pipeline focus on prediction, validation, and personalization, SCP focuses on decision support, ensuring that the system delivers practical value to the user. This layered integration aligns with modern intelligent system design,

where predictive analytics is combined with decision-support mechanisms to improve real-world outcomes [21].

#### 3.1.8.4 Technical Advantages and Contribution

The inclusion of the SCP module provides several important advantages that enhance the overall effectiveness of the proposed system. First, it bridges the gap between prediction and action by converting numerical outputs into intuitive and actionable recommendations. This significantly improves usability, as users can make informed decisions without interpreting complex data.

Second, SCP contributes directly to food waste reduction by guiding users toward timely consumption of perishable items. By prioritizing items based on urgency, the system ensures that resources are utilized efficiently and spoilage is minimized.

Third, the module enhances system interpretability by providing clear and understandable outputs. Unlike raw numerical predictions, priority levels are easy to interpret and act upon, increasing user engagement and trust.

Fourth, SCP supports scalability and extensibility. The priority scoring mechanism can be easily extended to incorporate additional factors such as nutritional value, user preferences, or consumption patterns, enabling future enhancements without significant architectural changes.

Finally, the integration of SCP with AEIF and AED results in a complete hybrid intelligent system that combines prediction, validation, personalization, and decision support within a unified framework. This comprehensive approach distinguishes the proposed system from existing solutions, which typically address only isolated aspects of the problem.

## 3.2 Tools and Technologies

The development and implementation of the proposed Personalized Food Expiry Prediction System required the integration of multiple tools and technologies spanning machine

learning, backend development, database management, and frontend user interaction. These technologies were selected to ensure scalability, efficiency, and seamless system integration.

The machine learning component of the system was developed using Python, which provides a robust ecosystem for data science and model development. Libraries such as Pandas and NumPy were used for data manipulation and preprocessing, enabling efficient handling of structured datasets. The predictive model was implemented using the CatBoost library, which is specifically designed for handling categorical features and structured tabular data. CatBoost's ability to model complex nonlinear relationships and reduce overfitting made it highly suitable for the food expiry prediction task [7].

For backend development, the system utilizes the Flask framework, a lightweight and flexible web framework that facilitates the creation of RESTful APIs. Flask enables seamless communication between the frontend interface and the machine learning model, allowing real-time prediction and data processing. The backend is responsible for handling user requests, processing input data, invoking the prediction pipeline (AEIF, AED, SCP), and returning results to the user interface.

The system employs MongoDB as its primary database for storing user data, food inventory records, prediction history, and feedback information. MongoDB's NoSQL structure allows flexible schema design, which is particularly useful for handling dynamic and evolving data such as user feedback and personalized adjustments. The integration of MongoDB with Flask through libraries such as Flask-PyMongo enables efficient data storage and retrieval operations.

On the frontend, the system is developed using React, a modern JavaScript library for building interactive user interfaces. React enables the creation of dynamic components that allow users to input food data, view predictions, and receive notifications. The frontend communicates with the backend through API calls, ensuring a responsive and user-friendly experience. Additional tools such as Axios are used for handling HTTP requests, while standard web technologies including HTML and CSS are used for layout and styling.

The overall system development process was supported by tools such as GitHub for version control and collaboration, ensuring proper tracking of code changes and team contributions. The development environment was managed using Python virtual environments to maintain dependency consistency across different modules.

Collectively, these tools and technologies enable the implementation of a full-stack intelligent system that integrates machine learning, data management, and user interaction within a unified platform. The selection of widely adopted and scalable technologies ensures that the system can be extended and deployed in real-world environments with minimal constraints.

### 3.3 Commercialization aspect of the product

The proposed Personalized Food Expiry Prediction System demonstrates strong commercialization potential due to its ability to address critical real-world challenges related to food waste reduction, cost efficiency, and intelligent food management. With increasing global concern over food sustainability and resource optimization, there is a growing demand for AI-driven solutions that provide accurate, actionable, and user-centric insights into food consumption [1]. Unlike conventional solutions, the proposed system integrates predictive intelligence, safety validation, personalization, and decision support within a unified framework, offering a comprehensive and practical solution for both individual and commercial users.

At the household level, the system can be commercialized as a web-based application that enables users to manage their food inventory efficiently. By providing personalized expiry predictions through the AEIF, along with adaptive learning AED and consumption prioritization SCP, the system delivers highly relevant and actionable insights tailored to individual usage patterns. This not only reduces food waste but also improves cost savings and convenience for users. The inclusion of notification systems and intuitive user interfaces further enhances accessibility and adoption potential among a broad consumer base.

Beyond individual usage, the system has significant applicability in retail and supply chain environments, including supermarkets, restaurants, and food distribution networks.

Businesses can leverage the system to monitor inventory in real time, predict spoilage risks, and optimize stock management strategies. By identifying items approaching expiry, organizations can implement dynamic pricing, targeted promotions, and redistribution strategies, thereby minimizing financial losses and improving operational efficiency. The integration of predictive analytics into supply chain workflows aligns with modern data-driven decision-making practices in the food industry.

Furthermore, the system can be extended into smart ecosystem integrations, including IoT-enabled kitchen appliances such as intelligent refrigerators and storage systems. By combining real-time sensor data with predictive modeling, the system can provide automated tracking and continuous expiry estimation without manual input. This aligns with emerging trends in smart home technologies, where AI-driven solutions enhance automation, efficiency, and user convenience.

From a business model perspective, the system supports multiple monetization strategies. These include subscription-based services for individual users, enterprise-level solutions for retail and logistics organizations, and API-based integration for third-party platforms such as grocery applications and smart home systems. The modular and scalable architecture of the system ensures adaptability across different market segments, enabling gradual expansion from consumer applications to large-scale commercial deployments.

Overall, the proposed system offers a unique combination of predictive accuracy, safety assurance, personalization, and decision support, positioning it as a viable and competitive solution in the emerging market of intelligent food management systems.

### 3.3.1 Future Scope

While the proposed system demonstrates strong performance and practical applicability, several opportunities exist for further enhancement and expansion. One key direction for future work is the integration of real-time IoT sensor data, such as temperature and humidity monitoring devices, to provide more accurate and continuous environmental inputs. This would enable the system to move beyond user-provided data and incorporate real-time tracking, further improving prediction accuracy.

Another potential enhancement is the incorporation of computer vision techniques for automated food recognition and spoilage detection. By utilizing image-based models, the system can identify food items and detect visual signs of degradation, reducing reliance on manual input and enhancing user convenience.

The personalization mechanism can also be extended through the use of advanced adaptive learning models, such as reinforcement learning or online learning algorithms, which can continuously update model parameters based on user interactions. This would enable more sophisticated and dynamic personalization capabilities.

Additionally, future work can explore the integration of nutritional analysis and dietary planning, linking expiry prediction with health recommendations. By combining food expiry data with nutritional requirements, the system can provide holistic guidance that supports both food waste reduction and health optimization.

From a scalability perspective, the system can be expanded to support multi-user environments and large-scale deployments, particularly in commercial settings such as supermarkets and supply chains. Enhancements in cloud-based infrastructure and distributed processing can further improve system performance and accessibility.

Finally, future research can focus on improving model interpretability and explainability, ensuring that users understand how predictions are generated and adjusted. This is particularly important for building trust in AI-driven systems, especially in domains involving safety and consumption decisions [25].

### 3.4 Testing and Implementation

The proposed system was rigorously tested to ensure its accuracy, reliability, and practical usability under real-world conditions. The testing process involved multiple stages, including model evaluation, functional testing, API validation, and real-world scenario testing.

### 3.4.1 System Implementation

The Personalized Food Expiry Prediction System was implemented as a full-stack intelligent application that integrates machine learning, rule-based validation, adaptive learning, and decision-support mechanisms within a unified architecture. The system is designed to transform real-world user inputs into reliable and actionable expiry predictions through a structured pipeline that combines predictive intelligence with domain knowledge and user feedback. The implementation consists of a React-based frontend for user interaction, a Flask-based backend for processing and orchestration, and a MongoDB database for persistent storage. The predictive intelligence is driven by a trained CatBoost regression model, which is tightly integrated with the AEIF, AED, and SCP modules.

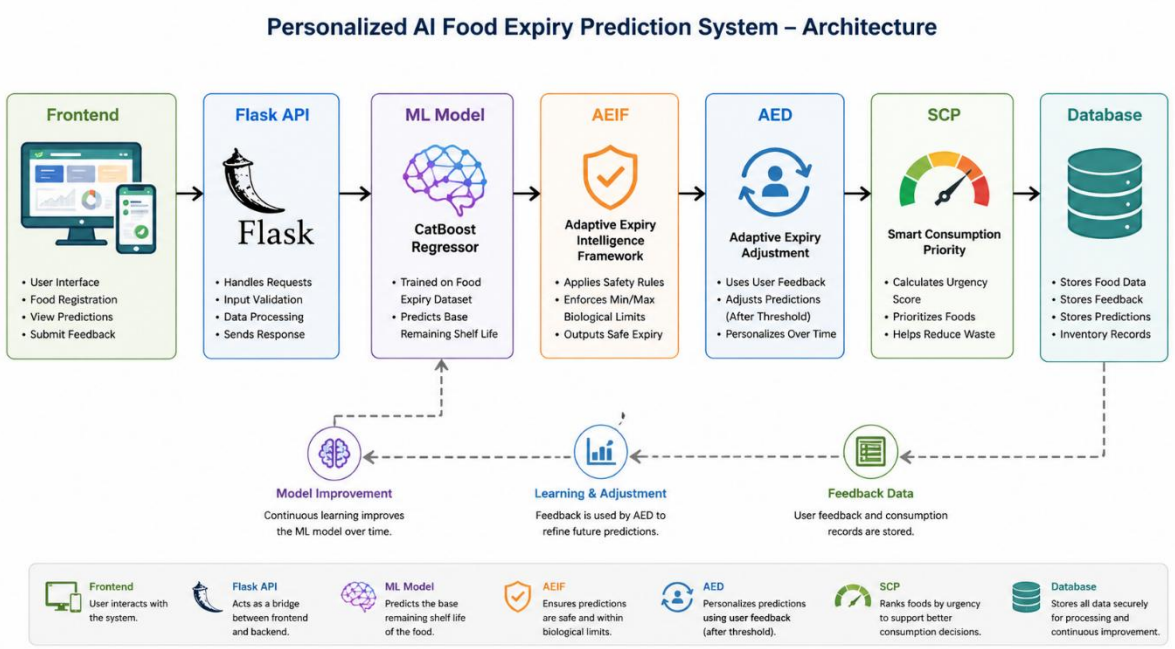


Fig. 3.6: Overall Implementation Architecture of the Proposed System

The machine learning model is trained offline and deployed as a pre-trained artifact, ensuring efficient runtime performance. During execution, the model is loaded from a serialized file and used exclusively for inference, eliminating the need for retraining during user interaction. To ensure consistency between training and inference, the system reconstructs the exact feature structure used during model training by loading a predefined feature column mapping.

User inputs such as food item, category, storage type, and environmental conditions are first validated and canonicalized to maintain consistency with the model's vocabulary. These inputs are then transformed into a structured feature vector using one-hot encoding, where all expected features are initialized and selectively activated based on the input data. Environmental variables such as temperature and humidity are incorporated into the feature set and are constrained within predefined bounds to prevent unrealistic or out-of-distribution predictions.

Once the feature vector is constructed, it is passed to the CatBoost model, which generates a baseline prediction representing the estimated number of days until expiry. However, this prediction is not directly exposed to the user. Instead, it is processed through the Adaptive AEIF, which enforces domain-specific safety constraints. The AEIF mechanism ensures that predicted values do not fall below a minimum safety threshold derived from base expiry knowledge and remain within biologically plausible limits. This step is essential for preventing unrealistic outputs that could arise from purely data-driven models, thereby enhancing the reliability and safety of the system.

Following safety validation, the system evaluates whether personalization should be applied through the AED module. As implemented in the backend logic , personalization is activated only when a minimum threshold of user feedback instances has been reached for a specific food item. This threshold-based mechanism ensures that sufficient data is available before applying adaptive adjustments, thereby preventing unstable or inaccurate personalization during early usage. When the threshold condition is satisfied, the AED module modifies the baseline prediction using learned adjustment parameters derived from historical user feedback. These adjustments are computed using an error-based learning approach, where the difference between predicted and actual expiry is used to refine future predictions. The learning process incorporates controlled update mechanisms, including learning rate scaling and decay factors, to ensure stability and gradual adaptation over time.

In addition to prediction and personalization, the system incorporates a decision-support layer through the SCP module. This component converts predicted expiry values into actionable insights by assigning priority scores based on the number of days remaining until expiry. Items

approaching their expiry date are assigned higher priority, enabling users to identify and consume them promptly. This prioritization mechanism enhances the practical usability of the system by bridging the gap between prediction and real-world decision-making.

The backend API acts as the central orchestration layer that integrates all components of the system. The prediction workflow is implemented within the `/predict` endpoint, where user inputs are processed through a sequential pipeline consisting of input validation, feature construction, model prediction, safety validation, personalization adjustment, and priority scoring. The final results, including baseline prediction, personalized prediction, final expiry date, remaining days, and priority score, are returned to the frontend interface for user display. At the same time, the system persists all relevant data in MongoDB, including prediction results, feedback records, adjustment parameters, and historical prediction logs. The implementation maintains a rolling prediction history for each food item, allowing the system to track changes over time and support analytical insights.

Overall, the implementation demonstrates a tightly integrated hybrid system where machine learning, domain knowledge, and adaptive user feedback operate cohesively. The design ensures that predictions are not only accurate but also safe, personalized, and actionable, thereby addressing the limitations of existing food expiry prediction systems. The combination of a robust backend architecture, efficient model deployment, and continuous feedback-driven adaptation results in a scalable and practical solution for real-world food management.

#### 3.4.1.1 Model Loading and Prediction Execution

The trained CatBoost model is loaded at runtime using the prediction module, ensuring that inference uses the exact trained configuration.

```
self.model = CatBoostRegressor()

self.model.load_model(MODEL_PATH)
```

This snippet shows that the model is not retrained during execution but instead loaded from a pre-trained `.cbm` file, ensuring efficiency and consistency. Additionally, the system loads a predefined feature structure:

```
with open(FEATURE_PATH, "r", encoding="utf-8") as f:
    self.feature_columns = [line.strip() for line in f.readlines()]
```

This guarantees that the input feature order during prediction matches the training phase, which is critical for correct model behavior.

#### 3.4.1.2 Feature Engineering and Input Construction

Before prediction, user inputs are transformed into a structured feature vector:

```
row = {col: 0 for col in self.feature_columns}
```

This ensures all expected features exist. One-hot encoding is applied dynamically:

```
food_col = f"food_{item_name}"
if food_col in row:
    row[food_col] = 1
```

Environmental conditions are also incorporated:

```
temp_c = self._clamp(temp, -30.0, 40.0)
hum_p = self._clamp(hum, 40.0, 95.0)
```

This step ensures that input values remain within realistic bounds, preventing out-of-distribution predictions.

#### 3.4.1.3 AEIF Safety Validation

After generating raw predictions, the system enforces safety constraints:

```
min_safe = 0.60 * base_days
final_days = max(raw_pred_days, min_safe)
```

This ensures predictions never fall below 60% of the base expiry value. Additionally:

```
final_days = max(final_days, 1.0)
final_days = min(final_days, MAX_EXPIRY_DAYS)
```

This prevents unrealistic outputs such as zero-day expiry or excessively large predictions. This implementation demonstrates how domain knowledge is integrated into machine learning predictions.

#### 3.4.1.4 Personalization Activation (AED Logic)

A key innovation in the system is controlled personalization. The activation condition is implemented as:

```
MIN_FEEDBACK_FOR_PERSONALIZATION = 5

personalization_enabled = item_feedback_count >=
MIN_FEEDBACK_FOR_PERSONALIZATION
```

This ensures that personalization is only applied when sufficient feedback is available.

When enabled, personalization modifies predictions:

```
personalized_days = apply_aed(user_aed, item_name, category,
baseline_days, base_days)
```

This adjustment is based on historical user feedback, making predictions adaptive and user-specific.

#### 3.4.1.5 Prediction Pipeline Integration (Backend Flow)

The complete prediction pipeline is implemented in the `/predict` API:

```
ml = predictor.predict(data)

baseline_days = float(ml["final_days_until_expiry"])
```

Followed by personalization:

```
if personalization_enabled:

    personalized_days = apply_aed(...)
```

Then safety cap using printed expiry:

```
if printed_dt and model_dt > printed_dt:

    model_dt = printed_dt
```

Finally, SCP priority is calculated:

```
scp = float(scp_score(days_left))
```

This sequence clearly shows how multiple layers (ML → AEIF → AED → SCP) are integrated into a single pipeline.

#### 3.4.1.6 Data Persistence and History Tracking

The system stores predictions and history in MongoDB:

```
foods_col.update_one(

    {"_id": ObjectId(food_id)},

    {"$push": {"predictionHistory": {"$each": [history_entry], "$slice": -
20}}}

)
```

This ensures:

- Only last 20 predictions are stored
- Historical analysis is possible
- Personalization evolution can be tracked

#### 3.4.1.7 User Interface Implementation

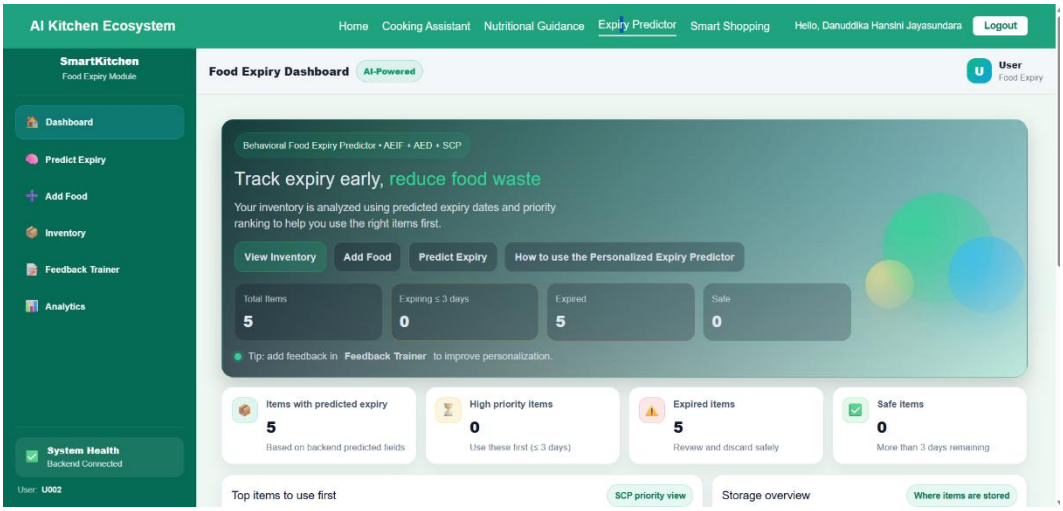


Fig 3.7: User Dashboard Interface

The dashboard provides a high-level overview of the user's food inventory, including total items, expired items, and priority-based insights. It integrates the SCP mechanism to highlight urgent items and supports navigation to core functionalities such as prediction, inventory management, and feedback training. This interface demonstrates how predictive outputs are translated into actionable insights for users.

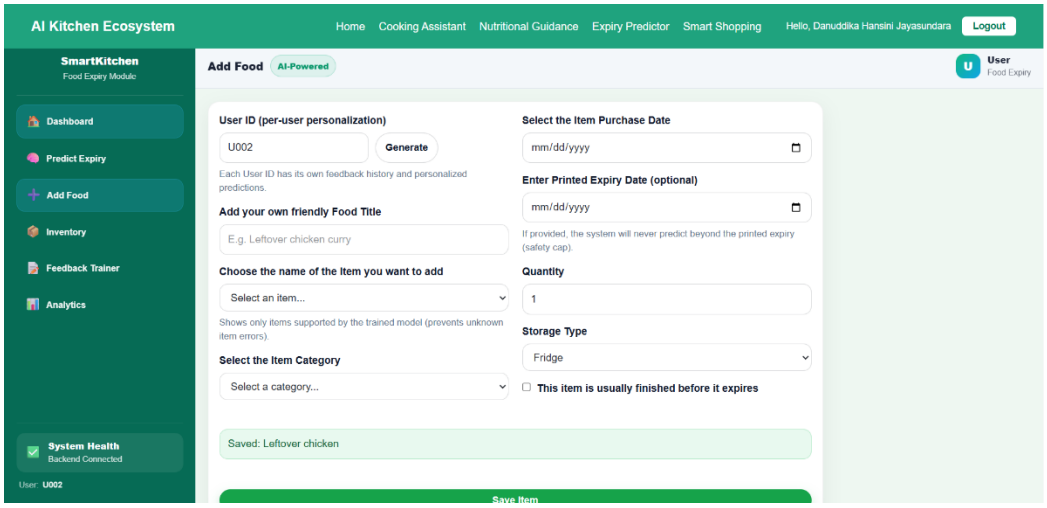


Fig 3.8: Food Registration Interface

This interface enables users to input food-related data, including item name, category, storage type, purchase date, and optional printed expiry date. It ensures structured data collection required for model prediction while supporting per-user personalization through unique user identifiers. This step forms the input layer of the prediction pipeline.

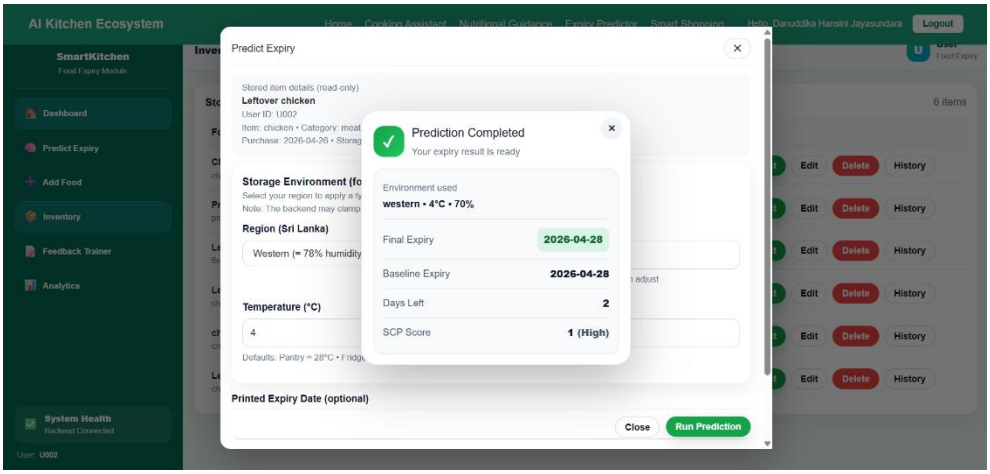


Fig 3.9: Expiry Prediction Output Interface

The prediction interface displays the final expiry date, baseline prediction, remaining days, and SCP priority score. The output reflects the integration of the machine learning model with AEIF safety validation and SCP decision support. This interface provides users with clear, interpretable results that support informed food consumption decisions.

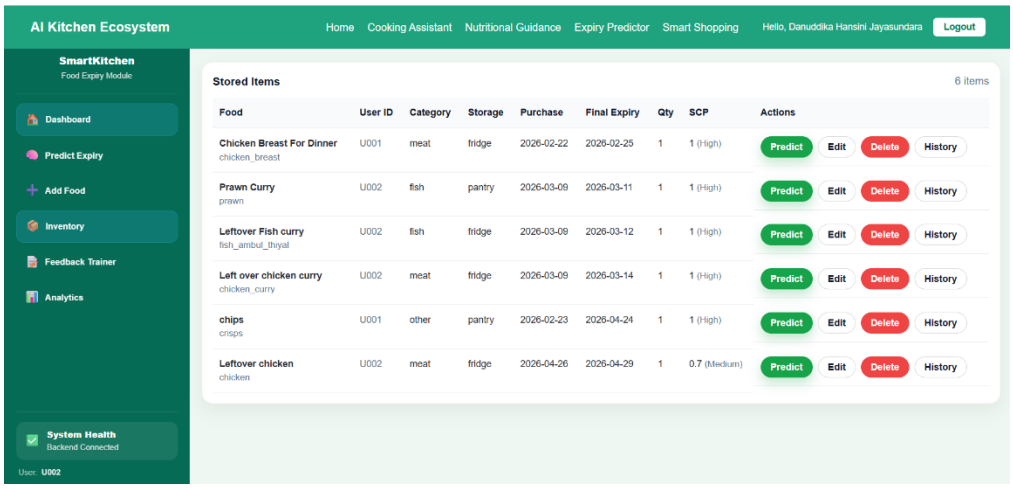


Fig 3.10: Inventory Management Interface

The inventory interface presents stored food items along with predicted expiry dates and assigned SCP priority levels. It allows users to perform actions such as prediction, editing, deletion, and viewing prediction history. This interface demonstrates how the system maintains and manages real-time food data with integrated decision-support features.

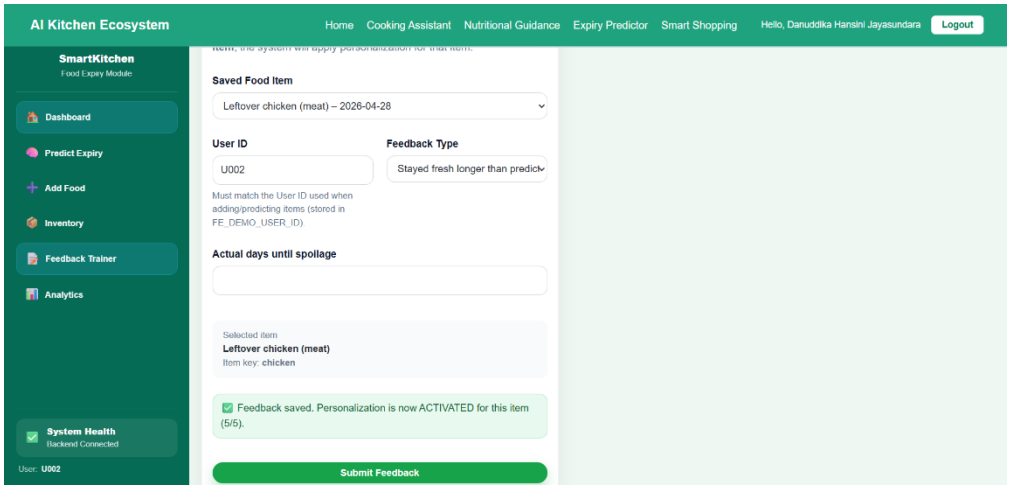


Fig 3.11: Feedback-Based Personalization Interface

This interface enables users to provide feedback on predicted expiry outcomes, including actual spoilage duration. Once sufficient feedback is collected, the AED module is activated, allowing the system to adjust predictions based on user-specific behavior. This component is critical for enabling adaptive learning and personalization.

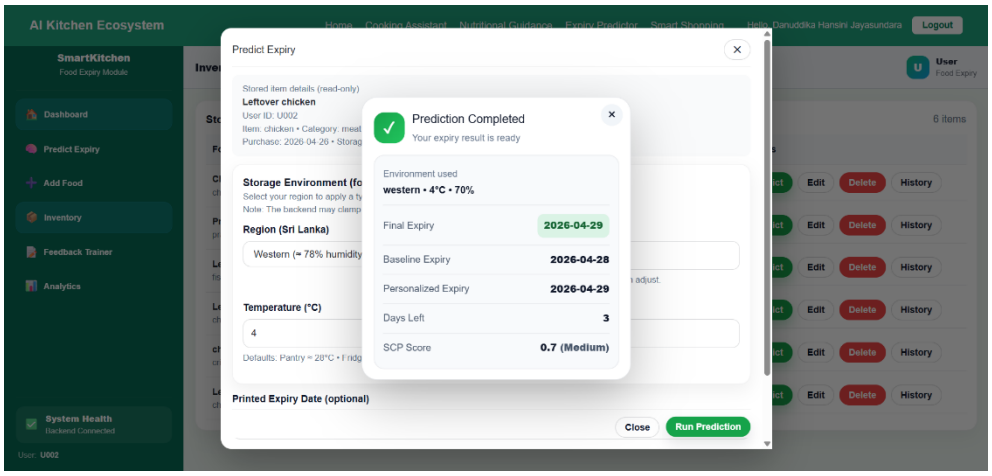


Fig 3.12: Personalized Expiry Prediction Output Interface

The prediction output interface presents the complete expiry estimation pipeline, including baseline prediction, personalized adjustment, and final expiry result. The baseline expiry represents the model's initial prediction, while the personalized expiry reflects adjustments applied through the AED mechanism based on user feedback. The final expiry value is produced after applying safety constraints defined in the AEIF. Additionally, the interface displays the remaining days and SCP score, enabling users to make informed and timely consumption decisions. This figure demonstrates the successful integration of machine learning, safety validation, and adaptive personalization within a unified system.

### 3.4.2 Testing

#### 3.4.2.1 Functional Testing

Functional testing was conducted to verify that all system components operate as expected. The prediction module was tested to ensure that it correctly generates expiry estimates based on user input. The AEIF layer was validated to confirm that safety constraints are applied correctly, preventing unrealistic predictions. The AED module was tested by simulating user feedback and verifying that predictions are adjusted accordingly. Similarly, the SCP module was evaluated to ensure accurate assignment of priority levels based on remaining shelf life.

Table 2: System Testing

| Test ID | Component Tested | Input Scenario | Expected Output | Result |
|---|---|---|---|---|
| TC01 | Food Addition | Valid food data | Stored in DB | Pass |
| TC02 | Prediction | Valid input | Expiry days returned | Pass |
| TC03 | AEIF Safety | Low prediction | Adjusted ≥60% base | Pass |
| TC04 | AED Activation | Feedback < 5 | No personalization | Pass |
| TC05 | AED Activation | Feedback ≥ 5 | Personalization applied | Pass |
| TC06 | SCP Ranking | Days ≤ 2 | High priority | Pass |

#### 3.4.2.2 API and Integration Testing

All backend API endpoints were tested using both the frontend interface and API testing tools. The system was validated to ensure that correct outputs are returned for valid inputs, while appropriate error handling mechanisms were triggered for invalid or incomplete inputs. Integration testing was also performed to confirm seamless communication between the frontend, backend, machine learning model, and database.

#### 3.4.2.3 Scenario-Based Testing

To evaluate real-world applicability, scenario-based testing was conducted. For example, a user adds a food item with specific storage conditions, and the system predicts its expiry duration. If the user later provides feedback indicating that the item expired earlier than predicted, the AED module adjusts future predictions accordingly. Such scenarios confirm that the system can adapt to real user behavior and improve accuracy over time.

#### 3.4.2.4 Edge Case Testing

Edge cases were tested to ensure system robustness under unusual conditions. These included missing environmental inputs, absence of printed expiry dates, and extreme temperature or humidity values. The system successfully handled these scenarios by applying default values or safety constraints, ensuring stable and reliable performance.

#### 3.4.2.5 User Interface Validation

The frontend interface was tested to ensure usability and clarity. Users were able to input data, view predictions, and understand priority levels without ambiguity. Visual indicators such as priority labels and notifications were verified to ensure effective communication of system outputs.

# 4. RESULTS AND DISCUSSION

## 4.1 Results

The performance of the proposed Personalized Food Expiry Prediction System was evaluated through a comprehensive analysis combining exploratory data analysis, statistical relationships, and predictive model evaluation. The results demonstrate not only the effectiveness of the machine learning model but also the importance of integrating environmental factors and domain-aware constraints within the prediction pipeline.

The exploratory analysis provides critical insights into how storage conditions influence food expiry behavior. As illustrated in Fig. 4.1, the distribution of expiry duration across different storage types reveals a clear and consistent trend. Food items stored in freezer conditions exhibit the highest median expiry duration, followed by refrigerator storage, while pantry storage shows the lowest median values. In addition to median differences, the interquartile range (IQR) for freezer storage is significantly wider, indicating that while freezing generally extends shelf life, the extent of preservation varies depending on the food type. Pantry storage, in contrast, shows higher dispersion and several extreme outliers, reflecting the mixed nature of items stored at room temperature, ranging from highly perishable goods to long-lasting dry

products. This variability highlights the necessity of incorporating both storage type and food category into the predictive model to avoid generalized estimations.

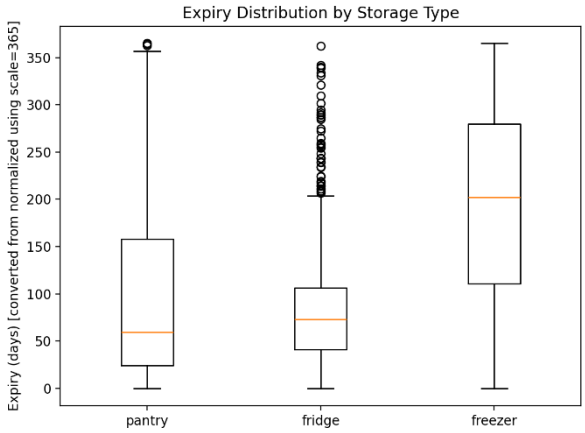


Figure 4.1: Expiry Distribution by Storage Type

The category-wise distribution of expiry duration, presented in Fig. 4.2, further emphasizes the heterogeneity of food characteristics. Categories such as meat and dairy display lower median expiry values and tighter distributions, indicating rapid and relatively predictable spoilage behavior. In contrast, categories like grains and snacks exhibit significantly higher median values and broader distributions, suggesting longer shelf life and greater tolerance to environmental variations. Notably, fruits and vegetables show moderate expiry durations but with considerable variability, reflecting their sensitivity to both storage conditions and intrinsic biological factors. The presence of outliers across multiple categories indicates that certain items within the same category may behave differently under varying conditions, reinforcing the need for a data-driven approach rather than relying solely on fixed expiry rules.

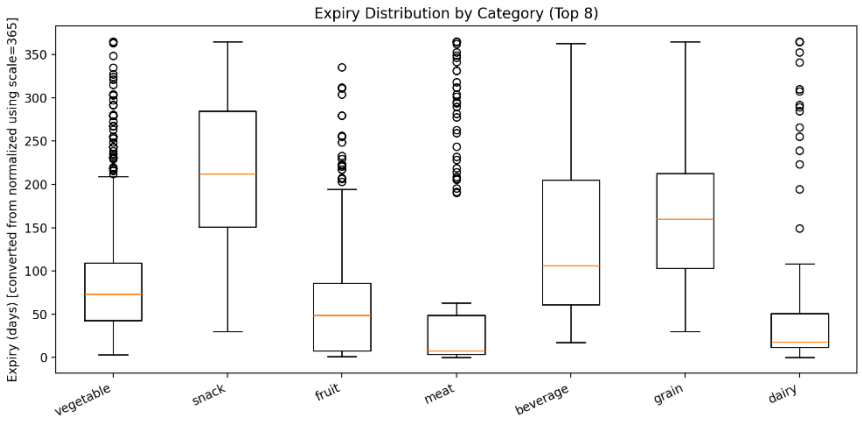

Figure 4.2: Expiry Distribution by Category

Environmental variables were further analyzed to understand their direct influence on food degradation. As shown in Fig. 4.3, the relationship between temperature and expiry duration demonstrates a clear negative trend, with a correlation coefficient of approximately -0.276. Although the correlation is moderate, the scatter plot reveals a distinct pattern where higher temperatures are associated with shorter expiry durations. The clustering of data points at lower temperatures (e.g., freezer and refrigerator ranges) with higher expiry values confirms the effectiveness of cold storage in prolonging shelf life. Additionally, the spread of points at higher temperatures indicates increased variability and uncertainty in expiry prediction under ambient conditions. This behavior justifies the inclusion of temperature as a critical feature in the predictive model.

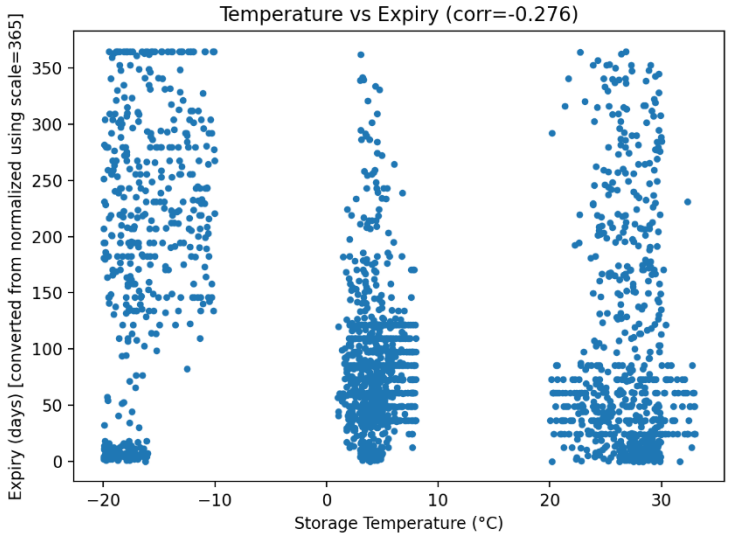


Figure 4.3: Temperature Vs Expiry

In contrast, Fig. 4.4 illustrates the relationship between humidity and expiry duration, where the correlation is weak (approximately -0.044). The scatter distribution shows no clear linear trend, suggesting that humidity alone is not a strong predictor of expiry duration within the dataset. However, the wide spread of data points indicates that humidity may still have a

secondary or interaction effect when combined with other variables such as temperature and storage type. Therefore, while humidity is included as a feature, its influence is comparatively less significant than temperature, aligning with observed statistical patterns.

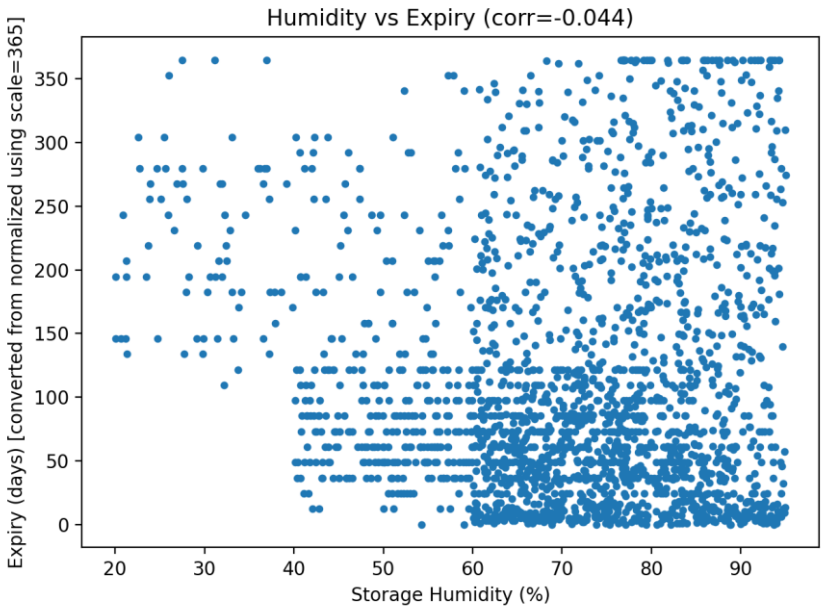


Figure 4.4: Humidity Vs Expiry

The effectiveness of the safety validation mechanism within the Adaptive Expiry Intelligence Framework is demonstrated in Fig. 4.5, which compares predicted expiry values against base expiry references. The distribution of predicted values around the base expiry line indicates that the model produces outputs that are closely aligned with known reference values while still allowing flexibility for environmental and contextual adjustments. Importantly, the absence of extreme deviations confirms that the safety constraints successfully prevent unrealistic predictions, ensuring that outputs remain within biologically acceptable limits.

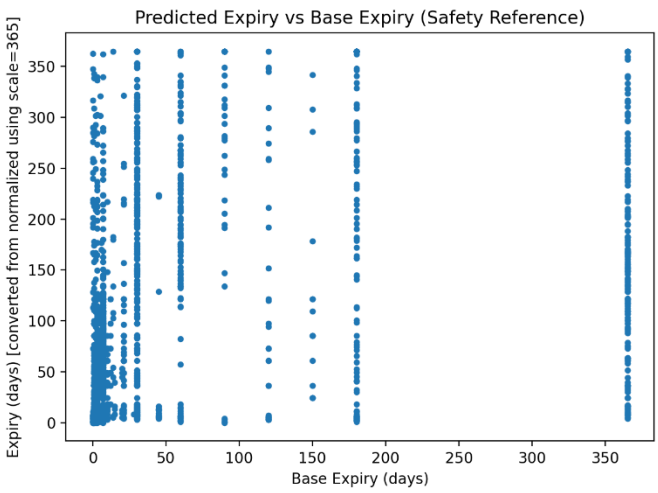


Figure 4.5: Predicted Expiry Vs Base Expiry

Further validation is provided by Fig. 4.6, which presents the distribution of the ratio between predicted expiry and base expiry values. The concentration of values around a ratio of 1 indicates that the majority of predictions are consistent with base expectations, while slight deviations reflect the influence of environmental conditions and learned patterns. The limited spread and absence of extreme outliers demonstrate the stability of the prediction system and the effectiveness of the safety bounding mechanism in maintaining controlled variability.

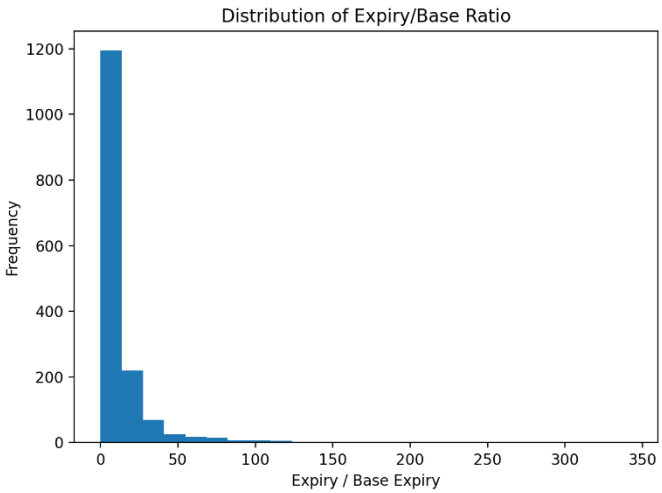


Figure 4.6: Distribution of Expiry

The predictive performance of the system was evaluated using multiple regression models to identify the most suitable approach. Among the evaluated models, the CatBoost regression model achieved the highest performance, with an $R^2$ score of 0.7443 and a Mean Absolute Error (MAE) of 0.096 days. This indicates that the model is capable of explaining approximately 74% of the variance in expiry duration while maintaining a very low prediction error. Compared to other models, CatBoost demonstrates superior capability in handling structured tabular data and categorical features, which are critical in this domain. These results confirm that the selected model, combined with the AEIF framework, provides a reliable and accurate solution for food expiry prediction.

## 4.2 Research Findings

The experimental evaluation of the proposed Personalized Food Expiry Prediction System led to several significant findings that validate both the modeling approach and the overall system design. These findings highlight the importance of integrating environmental factors, food characteristics, and adaptive intelligence within a unified predictive framework.

A primary finding of this study is that storage conditions are a dominant factor influencing food expiry duration. The clear separation observed in Fig. 4.1 between freezer, refrigerator, and pantry storage confirms that temperature-controlled environments significantly extend shelf life. Freezer storage consistently demonstrates the highest median expiry duration, while pantry storage exhibits both lower median values and higher variability. This indicates that storage type alone introduces substantial variation in expiry behavior, reinforcing the necessity of incorporating it as a key feature in the predictive model. Without this differentiation, predictions would lack contextual accuracy and fail to reflect real-world conditions.

A second important finding is that food category plays a critical role in determining expiry patterns. As observed in Fig. 4.2, perishable categories such as meat and dairy show consistently lower expiry durations, while grains and snack items exhibit significantly longer shelf lives. Additionally, categories such as fruits and vegetables display moderate but highly variable expiry durations, indicating sensitivity to both intrinsic biological properties and

external environmental conditions. This variability demonstrates that expiry prediction cannot rely solely on environmental factors and must account for the inherent characteristics of different food types. The effectiveness of the CatBoost model in handling categorical variables further supports this observation.

Another key finding relates to the impact of environmental variables on food degradation. The analysis presented in Fig. 4.3 confirms that temperature has a measurable and meaningful influence on expiry duration, with a moderate negative correlation (approximately -0.276). This indicates that as storage temperature increases, the rate of spoilage accelerates, reducing shelf life. In contrast, humidity, as shown in Fig. 4.4, exhibits only a weak correlation with expiry duration (approximately -0.044), suggesting that its independent effect is limited. However, the dispersion of data points implies that humidity may contribute indirectly when interacting with other variables. This finding emphasizes the importance of feature selection and prioritization, where temperature is treated as a primary factor while humidity serves as a supporting variable.

A further critical finding is the effectiveness of the Adaptive Expiry Intelligence Framework (AEIF) in ensuring prediction safety and stability. The relationship between predicted expiry and base expiry values, illustrated in Fig. 4.5, demonstrates that predictions are consistently aligned with known reference values while remaining within controlled bounds. The absence of extreme deviations confirms that the safety validation layer successfully constrains predictions, preventing unrealistic outputs. This is further supported by Fig. 4.6, where the distribution of the expiry-to-base ratio is concentrated around unity, indicating that the system maintains a balance between predictive flexibility and domain-based constraints. This finding validates the core contribution of the proposed framework in addressing a key limitation of traditional machine learning models, which often lack safety-awareness.

The comparative evaluation of machine learning models reveals that ensemble-based methods outperform both linear and neural approaches in this context. The CatBoost model achieved the highest predictive performance, followed closely by Random Forest and XGBoost, while Linear Regression and Neural MLP showed comparatively lower accuracy. This finding highlights the importance of selecting models that are well-suited for structured tabular data

with mixed feature types. The superior performance of CatBoost can be attributed to its ability to efficiently handle categorical variables and capture complex nonlinear interactions, making it particularly effective for food expiry prediction.

Finally, the integration of adaptive and decision-support components leads to the finding that prediction accuracy alone is insufficient without usability and contextual intelligence. The incorporation of the AED module enables the system to learn from user feedback and refine predictions over time, while the SCP module translates predictions into actionable insights. Together, these components transform the system from a static predictive model into a dynamic, user-centric decision-support system. This holistic approach enhances both practical applicability and user engagement, demonstrating the value of combining prediction, validation, adaptation, and prioritization within a single framework.

Overall, these findings confirm that accurate and reliable food expiry prediction requires a multi-dimensional approach that integrates environmental factors, food characteristics, machine learning techniques, and adaptive system intelligence. The proposed system successfully captures these dimensions, resulting in a robust and practical solution for real-world food management challenges.

### 4.3 Model Evaluation

Table 3: Model Comparison

| Model | MAE | $R^2$ Score | Remarks |
|---|---|---|---|
| CatBoost | 0.0960 | 0.7443 | Best performance |
| Random Forest | 0.0968 | 0.7379 | Slightly lower |
| XGBoost | 0.0993 | 0.7343 | Competitive |
| Neural MLP | 0.1318 | 0.5256 | Moderate |
| Linear Regression | 0.1459 | 0.4452 | Weak |

The results show that Linear Regression achieved the lowest performance, with an $R^2$ score of 0.4452 and MAE of 0.1459 days. This indicates that a simple linear model cannot

sufficiently capture the complex relationships between food type, storage method, temperature, humidity, and expiry duration. Food spoilage is influenced by nonlinear interactions; for example, the effect of temperature varies depending on whether the item is dairy, meat, grain, or fruit. Therefore, the weak performance of Linear Regression confirms that the expiry prediction task requires more advanced modeling techniques.

The MLP Regressor improved performance compared to Linear Regression, achieving an $R^2$ score of 0.5256 and MAE of 0.1324 days. This shows that the neural network was able to learn some nonlinear relationships in the dataset. However, its performance remained significantly lower than the tree-based ensemble models. This may be due to the structured tabular nature of the dataset and the high number of one-hot encoded categorical features. Neural networks generally require larger datasets and careful tuning to outperform ensemble tree models in tabular prediction tasks. Therefore, while the MLP served as a useful deep learning baseline, it was not selected as the final model.

The tree-based ensemble models produced much stronger results. XGBoost achieved an $R^2$ score of 0.7343 with an MAE of 0.0993 days, while Random Forest achieved an $R^2$ score of 0.7379 with an MAE of 0.0968 days. These results confirm that ensemble learning methods are well suited for food expiry prediction because they can capture nonlinear feature interactions and handle mixed data patterns more effectively than linear or neural approaches. The strong performance of both Random Forest and XGBoost also validates the quality of the engineered features, including storage type, food category, environmental values, and base expiry duration.

Among all evaluated models, CatBoost achieved the best performance, with the highest $R^2$ score of 0.7443 and the lowest MAE of 0.0960 days. Although the improvement over Random Forest and XGBoost is small, CatBoost still produced the most accurate and consistent results. This is important because even small improvements in expiry prediction can affect user trust and practical decision-making in food management. CatBoost was particularly suitable for this system because the dataset contains many categorical variables, including food item names, categories, and storage types. Its ability to handle categorical and

tabular data effectively allowed it to model expiry behavior more accurately than the competing models.

Overall, the model evaluation demonstrates a clear performance progression from simple models to advanced ensemble methods. Linear Regression and MLP showed limited to moderate performance, while Random Forest, XGBoost, and CatBoost achieved substantially stronger results. The final selection of CatBoost is therefore justified by both its quantitative performance and its technical suitability for structured food expiry data. These findings confirm that the proposed system benefits from a robust tree-based boosting model as the core predictive engine, while AEIF, AED, and SCP further enhance safety, personalization, and decision support.

### 4.4 Discussions

The results of this study demonstrate that the proposed Personalized Food Expiry Prediction System effectively addresses the limitations of existing approaches by integrating predictive modeling with safety validation, adaptive learning, and decision-support mechanisms. Unlike traditional systems that rely either on static rule-based estimations or standalone machine learning models, the proposed framework combines multiple layers of intelligence to achieve both predictive accuracy and practical reliability.

One of the most significant observations from the results is the strong influence of storage conditions on food expiry behavior. The clear differentiation in expiry distributions across freezer, refrigerator, and pantry environments confirms that temperature-controlled storage plays a critical role in preserving food quality. This finding aligns with established food science principles, where lower temperatures inhibit microbial activity and enzymatic reactions responsible for spoilage. The incorporation of storage type and temperature into the predictive model therefore represents a necessary and effective design choice, ensuring that predictions reflect real-world conditions rather than generalized assumptions.

The analysis also highlights the importance of food category in determining expiry patterns. The observed differences between highly perishable categories such as meat and dairy and more stable categories such as grains and snacks demonstrate that food degradation is

inherently dependent on biological composition. This supports the use of categorical-aware machine learning models, particularly CatBoost, which is capable of capturing complex interactions between categorical and numerical features. The superior performance of CatBoost compared to linear and neural models further reinforces the importance of selecting algorithms that are well-suited for structured tabular data.

A key contribution of this work is the introduction of the AEIF, which addresses a critical limitation of conventional machine learning models, lack of safety awareness. While data-driven models can achieve high accuracy, they may produce predictions that are unrealistic or unsafe when applied in real-world scenarios. The results demonstrate that AEIF successfully constrains predictions within biologically valid ranges, as evidenced by the controlled distribution of predicted values relative to base expiry references. This integration of domain knowledge with machine learning aligns with emerging research trends that emphasize the importance of hybrid intelligent systems for improving reliability and trust in AI-based applications.

In addition to safety validation, the inclusion of the AED module introduces a dynamic learning capability that enhances system adaptability. By incorporating user feedback into the prediction process, the system evolves over time and aligns more closely with individual usage patterns. This represents a shift from static prediction models to adaptive systems that continuously improve based on real-world interactions. Such feedback-driven learning mechanisms are increasingly recognized as essential for building intelligent systems that operate effectively in dynamic environments.

The SCP module further strengthens the system by transforming predictive outputs into actionable recommendations. While many existing systems focus solely on prediction, they often fail to provide meaningful guidance to users. The SCP module addresses this gap by prioritizing food items based on urgency, enabling users to make informed decisions and reduce food waste. This integration of prediction and decision support reflects a holistic approach to problem-solving, where the system not only analyzes data but also facilitates practical action.

Despite these strengths, certain limitations must be acknowledged. The system currently relies on user-provided environmental inputs such as temperature and humidity, which may introduce inaccuracies if not measured precisely. Additionally, the dataset used for training, although comprehensive, may not fully capture all variations in food handling practices and regional conditions. The absence of real-time sensor data and visual inspection mechanisms also limits the system's ability to detect dynamic changes in food quality.

These limitations provide opportunities for future improvements. The integration IoT devices can enable real-time monitoring of environmental conditions, improving prediction accuracy and reducing reliance on manual input. Similarly, the incorporation of computer vision techniques can allow automated detection of food spoilage based on visual cues. Expanding the dataset to include more diverse food items and regional variations can further enhance model generalization.

Overall, the findings confirm that the proposed system provides a comprehensive and effective solution for food expiry prediction. By combining machine learning, safety validation, adaptive learning, and decision support within a unified framework, the system addresses both technical and practical challenges associated with food management. This multi-layered approach not only improves prediction accuracy but also enhances usability and real-world applicability, positioning the system as a significant advancement in intelligent food management solutions.

### 4.6 Ethical, Security and Limitations

The deployment of intelligent food expiry prediction systems introduces several ethical, security, and practical considerations that must be carefully addressed to ensure responsible and reliable usage. While the proposed system enhances food management through predictive analytics and adaptive learning, it operates in a domain that directly influences user health and consumption decisions. Therefore, ensuring transparency, safety, and user awareness is critical.

From an ethical perspective, the system provides predictions based on data-driven models and domain-specific constraints; however, it does not guarantee absolute accuracy in all

scenarios. Food spoilage is influenced by numerous external factors that may not be fully captured by the model, such as contamination, improper handling, or unexpected environmental fluctuations. As a result, there exists a risk that users may overly rely on the system's predictions when making consumption decisions. To mitigate this, the system is designed as a decision-support tool rather than an authoritative source, and users are implicitly encouraged to apply personal judgment and sensory evaluation when assessing food safety. This approach aligns with ethical guidelines for AI systems, where transparency and human oversight are essential in safety-critical applications.

In terms of security, the system handles user-generated data, including food inventory details, consumption behavior, and feedback records. Although the data does not typically include highly sensitive personal information, it is still important to ensure that it is stored and processed securely. The backend implementation incorporates input validation mechanisms to prevent malformed or malicious data from affecting system behavior. Additionally, the use of structured database management ensures data consistency and integrity. For large-scale deployment, further security enhancements such as authentication mechanisms, encrypted communication channels (HTTPS), and role-based access control can be integrated to protect user data and prevent unauthorized access.

The system also has a positive social impact, particularly in addressing the global issue of food waste. By providing accurate and personalized expiry predictions, the system encourages users to optimize food consumption and reduce unnecessary disposal. This contributes to both economic savings at the household level and environmental sustainability by minimizing food waste. Furthermore, the integration of prioritization mechanisms supports better decision-making, enabling users to manage their food resources more efficiently.

Despite its strengths, the proposed system has several limitations that must be acknowledged. First, the accuracy of the machine learning model is dependent on the quality and diversity of the training dataset. The dataset used in this study, while comprehensive, may not fully represent all possible food types, storage conditions, or regional variations. As a result, predictions for unseen or uncommon items may be less accurate. Second, the system

relies on predefined environmental assumptions when user inputs are incomplete. While default temperature and humidity values improve usability, they may not always reflect actual storage conditions, potentially affecting prediction accuracy.

Another limitation relates to the personalization mechanism implemented through the AED module. Personalization is activated only after a minimum threshold of user feedback is reached, ensuring stability in learning. However, this also means that new users initially receive generalized predictions without personalization. Although this is necessary to maintain reliability, it introduces a delay in achieving fully personalized performance. Additionally, the system does not currently incorporate real-time sensor data, such as IoT-based environmental monitoring, which could further enhance prediction accuracy by capturing dynamic storage conditions.

In summary, while the proposed system demonstrates strong capabilities in predictive accuracy, safety validation, and adaptive learning, it must be understood within the context of its ethical responsibilities and practical constraints. By acknowledging these considerations and designing the system with transparency, security, and user awareness in mind, the solution provides a balanced and responsible approach to intelligent food expiry prediction.

## 5. Conclusion

This research presented the design and development of a comprehensive Personalized Food Expiry Prediction System that integrates machine learning, safety validation, adaptive learning, and decision-support mechanisms to address the critical global issue of food waste. The study was motivated by the limitations of existing approaches, which either rely on static rule-based estimations or purely data-driven models that lack practical reliability and safety awareness. To overcome these challenges, a novel hybrid framework was proposed, combining predictive intelligence with domain-specific constraints and user-driven adaptation.

At the core of the system is the CatBoost regression model, which demonstrated strong predictive performance in estimating the remaining shelf life of food items. The model achieved an $R^2$ score of 0.7443 and a MAE of 0.096 days, indicating high accuracy and effective modeling of nonlinear relationships between food characteristics, environmental conditions, and expiry duration. Comparative evaluation with baseline models, including Linear Regression, Neural Networks, Random Forest, and XGBoost, confirmed the superiority of CatBoost for structured tabular data with categorical features.

A key contribution of this research is the introduction of the AEIF, which enhances prediction reliability by integrating safety validation mechanisms into the prediction pipeline. By constraining model outputs within biologically acceptable limits using base expiry references and safety thresholds, AEIF ensures that predictions are both accurate and safe for real-world usage. This addresses a critical limitation of conventional machine learning approaches, which may produce unrealistic or unsafe predictions when applied in practical contexts.

In addition to safety validation, the system incorporates the AED module, which enables continuous learning from user feedback. This personalization mechanism allows the system to adapt to individual usage patterns and environmental variations, improving long-term prediction accuracy and relevance. Furthermore, the SCP module translates predicted expiry values into actionable insights by assigning priority levels based on urgency, thereby supporting informed decision-making and reducing food waste.

The results of this study demonstrate that food expiry prediction is inherently influenced by multiple factors, including storage conditions, food category, and environmental variables. The integration of these factors within a unified framework significantly improves prediction accuracy and system usability. The proposed system not only provides reliable expiry predictions but also enhances practical applicability through safety assurance, personalization, and decision support, distinguishing it from existing solutions that address only isolated aspects of the problem.

Despite its strengths, the study acknowledges certain limitations, including reliance on user-provided environmental data and the absence of real-time monitoring mechanisms. These

limitations highlight opportunities for future work, such as the integration of IoT-based sensors for real-time data acquisition and the use of computer vision techniques for automated food condition assessment. Expanding the dataset to include a wider range of food items and regional variations can further improve model generalization and robustness.

In conclusion, this research demonstrates that a hybrid intelligent approach combining machine learning, domain knowledge, and adaptive learning can effectively address the challenges of food expiry prediction. The proposed system provides a scalable, practical, and user-centric solution that has the potential to significantly reduce food waste and improve food management practices. By bridging the gap between predictive accuracy and real-world usability, this work contributes to the advancement of intelligent food systems and opens new avenues for future research and application in sustainable food management.

# 5. DESCRIPTION OF PERSONNEL AND FACILITIES

## 5.1 Supervisor

Mrs. Lokesha Weerasinghe

Senior Lecturer

Faculty of Computing | Information Technology

Sri Lanka Institute of Information Technology

Malabe

## 5.2 Co-Supervisor

Mrs. Chathurya Kumarapperuma

Assistant Lecturer

Faculty of Computing | Information Technology

Sri Lanka Institute of Information Technology

Malabe

# 6. REFERENCES


[1] United Nations Environment Programme, *Food Waste Index Report 2024*. Nairobi, Kenya: UNEP, 2024.

[2] Food and Agriculture Organization of the United Nations, *World Food and Agriculture – Statistical Yearbook 2024*. Rome, Italy: FAO, 2024.

[3] D. Patra, M. Leisner, and A. M. C. Wong, "Understanding and addressing food waste due to date labeling confusion," *Waste Management Bulletin*, vol. 1, no. 2, pp. 100–108, 2022.

[4] M. Kavanaugh and J. J. Quinlan, "Consumer knowledge and behaviors regarding food date labels and food waste," *Food Control*, vol. 115, 2020.

[5] D. Li, Y. Zhang, and X. Chen, "Research progress of machine learning in extending and predicting food shelf life," *Foods*, vol. 13, no. 19, 2024.

[6] M. Rashvand, M. Alizadeh, and S. Hosseini, "Artificial intelligence for prediction of shelf-life of food products: A review," *Trends in Food Science & Technology*, 2025.

[7] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[8] F. Cui, X. Zhang, and Y. Li, "Machine learning-based shelf-life prediction for marine food under varying temperatures," *Food Chemistry*, 2024.

[9] H. S. El-Mesery, M. A. Hassan, and K. A. Al-Dhabi, "Artificial intelligence applications for predicting food shelf life and quality: A review," *Trends in Food Science & Technology*, 2025.

[10] K. Shehzad, "Predictive AI Models for Food Spoilage and Shelf-Life Estimation," *Global Trends in Science and Technology*, vol. 1, no. 1, pp. 75–94, 2025.

[11] J. Ruby Elizabeth, M. S. Kumar, and R. Prakash, “Machine Learning-Enhanced Shelf-Life Prediction Model for Packaged Foods Using Environmental Cues,” *International Journal of Advanced Signal and Image Sciences*, vol. 12, no. 1, pp. 262–273, 2026.

[12] J. Baranyi and T. A. Roberts, “A dynamic approach to predicting bacterial growth in food,” *International Journal of Food Microbiology*, vol. 23, no. 3–4, pp. 277–294, 1994.

[13] M. Ahmed and A. E. Hassanien, “Optimizing food quality prediction using machine learning techniques,” *Journal of Food Engineering*, 2022.

[14] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794.

[15] S. Nowshad and M. S. Khan, “Electronic tongue for food safety and quality assessment: A review,” *Food Chemistry*, 2021.

[16] Z. Zhang, Y. Liu, and X. Wang, “Intelligent discrimination of grain aging using machine learning techniques,” *Foods*, vol. 15, no. 2, 2026.

[17] G.-J. Nychas, E. Z. Panagou, and F. Mohareb, “Data science in the food industry,” *Annual Review of Food Science and Technology*, vol. 12, pp. 1–27, 2021.

[18] C. L. Arellano Vidal and J. E. Govan, “Machine learning techniques for improving nanosensors in agro-environmental applications,” *Agronomy*, vol. 14, no. 2, 2024.

[19] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, “CatBoost: unbiased boosting with categorical features,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[20] H. S. El-Mesery, M. A. Hassan, and K. A. Al-Dhabi, “Artificial intelligence applications for predicting food shelf life and quality: A review,” *Trends in Food Science & Technology*, 2025.

[21] R. Polikar, “Ensemble based systems in decision making,” *IEEE Circuits and Systems Magazine*, vol. 6, no. 3, pp. 21–45, 2006.

[22] L. A. Zadeh, “Fuzzy logic and approximate reasoning in decision systems,” *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 8, no. 4, pp. 249–257, 1978.

[23] G. B. Huang, Q. Y. Zhu, and C. K. Siew, “Extreme learning machine: Theory and applications,” *Neurocomputing*, vol. 70, no. 1–3, pp. 489–501, 2006.

[24] J. Fürnkranz and E. Hüllermeier, “Preference learning and ranking by pairwise comparison,” *Artificial Intelligence*, vol. 172, no. 16–17, pp. 1897–1916, 2008.

[25] P. Domingos, “A few useful things to know about machine learning,” *Communications of the ACM*, vol. 55, no. 10, pp. 78–87, 2012.

# 7. APPENDICES

## Appendix - A: Plagiarism Report

*Appendix 1 -- Plagiarism Report*